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PATIENT & SPECIMEN INFORMATION

Patient name	Hemingway, Ernest M.	**Accession**	NGS-2026-052101
Date of birth	1961-07-22 (Age 64)	**Specimen**	Oral swab, left lateral tongue
MRN	10345621	**Collected**	2026-05-19
Ordering physician	Dr. Wisdom, DDS MD	**Received**	2026-05-19
Clinical indication	Oral lesion — left lateral tongue	**Reported**	2026-05-20

RESULT: VARIANTS DETECTED

DETECTED VARIANTS

Gene	HGVS (coding)	Protein change	VAF	Depth	Tier	Classification
KRAS	NM_004985.5:c.35G>T	p.Gly12Asp	0.43%	4,630×	I	Pathogenic
PIK3CA	NM_006218.4:c.3140A>G	p.His1047Arg	0.70%	7,140×	I	Pathogenic

VARIANT ANNOTATION

KRAS p.Gly12Asp | Pathogenic (Tier I)

KRAS p.Gly12Asp (G12D) is one of the most frequently detected oncogenic driver mutations in human cancer. It locks KRAS in the GTP-bound active state, constitutively activating downstream RAS/MAPK and PI3K/AKT/mTOR proliferative signalling. In oral squamous cell carcinoma, KRAS mutations are found in 3–8% of cases and mark a subgroup with aggressive behaviour. This variant is classified Pathogenic (Tier I, COSMIC ID COSM521). The co-occurrence with PIK3CA H1047R (see below) indicates dual activation of RAS and PI3K pathways, a combination associated with resistance to single-agent targeted therapies and may influence enrolment eligibility in clinical trials. NOTE: Observed VAF (0.43%) is above the 0.2% reporting threshold; this call is supported by 2 high-quality consensus reads at 463× consensus depth. The low consensus read count (VD = 2) reflects limited target coverage at 2,000× raw input; orthogonal confirmation is recommended before clinical action.

PIK3CA p.His1047Arg | Pathogenic (Tier I)

PIK3CA p.His1047Arg is a hotspot activating mutation in the kinase domain of the p110α catalytic subunit of PI3-kinase. It constitutively activates the PI3K/AKT/mTOR pathway, promoting cell survival, proliferation, and tumour angiogenesis. PIK3CA mutations occur in 15–20% of OSCC, most commonly at E545K and H1047R hotspots. This variant is classified Pathogenic (Tier I, COSMIC ID COSM775). PIK3CA H1047R is potentially actionable: alpelisib (PI3Kα inhibitor) is FDA-approved for PIK3CA-mutated breast cancer and is under investigation in HNSCC. Multidisciplinary tumour board review for consideration of PI3K-directed therapy or clinical trial enrolment is recommended.

GENES WITH NO PATHOGENIC OR LIKELY PATHOGENIC VARIANTS DETECTED

AJUBA, CASP8, CDKN2A, CUL3, DPYD, EGFR, EP300, FAT1, FBXW7, HRAS, KEAP1, KMT2D, NFE2L2, NOTCH1, NSD1, PIK3R1, PTEN, TERT, TP53
Variants of uncertain significance (Tier III) are not reported.

METHOD SUMMARY

Cell-free and cellular DNA extracted from oral swab is subjected to hybridization capture using the IDT xGen Duplex UMI panel targeting 251 genomic regions (~77 kbp) across 21 cancer-relevant genes. Libraries incorporate 9-bp Unique Molecular Identifiers (UMIs) in a 9M1S+T read structure on both R1 and R2. Libraries are sequenced on Illumina NextSeq (2×150 bp). After alignment to GRCh38 (canonical chromosomes), reads are grouped by UMI using fgbio (adjacency strategy, ≤1 edit distance), and molecular consensus sequences are called requiring ≥2 reads per UMI family. Somatic variants are called on the consensus BAM with VarDict at minimum allele frequency 0.001. Variants ≥0.2% consensus VAF with ≥2 consensus supporting reads are reported. UMI deduplication reduces sequencing noise by >800-fold. Run-level negative controls are included on every sequencing run.

LIMITATIONS

This assay detects somatic single nucleotide variants and small indels within 251 target regions (~77 kbp) at VAF ≥0.2% with validated sensitivity of 98.4% at ≥20,000× raw input depth (~4,000× consensus after UMI deduplication). Structural variants, copy number alterations, gene fusions, large indels, intronic variants outside defined capture regions, and variants in regions with low hybridization capture efficiency are not reliably assessed. A negative result does not exclude malignancy. Tumour heterogeneity, low tumour cell content, and specimen quality may affect variant allele fraction. Variants below the 0.2% VAF reporting threshold may be present but are not reported. At 20,000× raw sequencing depth (~4,000× mean consensus), approximately 0.3% of targets may fall below 1,000× consensus depth; per-target coverage is provided in the laboratory technical report. Results should be interpreted in the context of clinical and pathological findings by a qualified clinician.

Reported by: Dr. Wisdom, MD PhD, FCAP **Report date: 2026-05-20**

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

HOTSPOT SCREENING SUMMARY

Patient: Hemingway, Ernest M. **Accession:** NGS-2026-052101

Hotspots listed below were explicitly interrogated in addition to all variants above 0.2 % VAF with ≥2 consensus reads across all 251 targets. Coverage criteria (≥100× consensus depth) were met at all targets.

EGFR		Cov ✓
L858R	c.2573T>G	Not detected
T790M	c.2369C>T	Not detected
Exon 19 del	c.2235_2249del	Not detected

KRAS		Cov ✓
G12D	c.35G>T	● **DETECTED**
G12V	c.35G>A	Not detected
G12C	c.34G>T	Not detected
G13D	c.38G>A	Not detected

PIK3CA		Cov ✓
E542K	c.1624G>A	Not detected
E545K	c.1633G>A	Not detected
H1047R	c.3140A>G	● **DETECTED**

TERT		Cov ✓
C228T	c.-124C>T	Not detected
C250T	c.-146C>T	Not detected

HRAS		Cov ✓
G12V	c.35G>T	Not detected
G12D	c.35G>A	Not detected
Q61L	c.182A>T	Not detected
Q61R	c.182A>G	Not detected

NOTCH1		Cov ✓
P2415L	c.7244C>T	Not detected
R1699W	c.5095C>T	Not detected

TP53		Cov ✓
R175H	c.524G>A	Not detected
R248Q	c.743G>A	Not detected
R248W	c.742C>T	Not detected
R273H	c.818G>A	Not detected
R282W	c.844C>T	Not detected

REMAINING PANEL GENES — SEQUENCE-LEVEL DETECTION, NO PREDEFINED HOTSPOT LIST

AJUBA · CASP8 · CDKN2A · CUL3 · DPYD · EP300 · FAT1 · FBXW7 · KEAP1 · KMT2D · NFE2L2 · NSD1 · PIK3R1 · PTEN

All variants above 0.2 % VAF with ≥2 consensus reads are reported for these genes. No variants meeting reporting criteria were detected. All targets in these genes met minimum coverage criteria (≥100× consensus depth).

Reported by: Dr. Wisdom, MD PhD, FCAP **Report date: 2026-05-20**

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

FULL PANEL COVERAGE REPORT — Hemingway, Ernest M. (NGS-2026-052101)

251 targets, 77,417 bp total. Mean consensus depth: 3,866×. Targets below 100× consensus: 0 / 251 (0%). Depths are derived from the consensus BAM (UMI deduplication, min-reads ≥ 2). Positions reported 1-based.

Gene	chr:start–end (1-based)	Len (bp)	Depth	Coverage status
DPYD				
DPYD	chr1:97,082,291–97,082,491	201	1,630×	✓ Covered
DPYD	chr1:97,449,958–97,450,158	201	275×	✓ Covered
DPYD	chr1:97,515,687–97,515,887	201	1,178×	✓ Covered
DPYD	chr1:97,573,763–97,573,963	201	1,857×	✓ Covered
PTEN				
PTEN	chr10:87,864,450–87,864,568	119	4,275×	✓ Covered
PTEN	chr10:87,894,005–87,894,129	125	9,773×	✓ Covered
PTEN	chr10:87,925,493–87,925,577	85	1,685×	✓ Covered
PTEN	chr10:87,931,026–87,931,109	84	7,231×	✓ Covered
PTEN	chr10:87,932,993–87,933,271	279	1,810×	✓ Covered
PTEN	chr10:87,952,098–87,952,279	182	2,142×	✓ Covered
PTEN	chr10:87,957,833–87,958,039	207	1,519×	✓ Covered
PTEN	chr10:87,960,874–87,961,138	265	926×	✓ Covered
PTEN	chr10:87,965,267–87,965,489	223	4,611×	✓ Covered
HRAS				
HRAS	chr11:532,619–532,775	157	3,113×	✓ Covered
HRAS	chr11:533,433–533,632	200	393×	✓ Covered
HRAS	chr11:533,746–533,964	219	2,463×	✓ Covered
HRAS	chr11:534,192–534,342	151	1,777×	✓ Covered
KRAS				
KRAS	chr12:25,209,778–25,209,931	154	1,099×	✓ Covered
KRAS	chr12:25,225,594–25,225,793	200	7,066×	✓ Covered
KRAS	chr12:25,227,214–25,227,432	219	559×	✓ Covered
KRAS	chr12:25,245,254–25,245,404	151	11,088×	● **VARIANT DETECTED**
KMT2D				
KMT2D	chr12:49,021,763–49,021,892	130	1,369×	✓ Covered
KMT2D	chr12:49,022,023–49,022,171	149	6,775×	✓ Covered
KMT2D	chr12:49,022,260–49,022,373	114	2,960×	✓ Covered
KMT2D	chr12:49,022,570–49,022,895	326	3,909×	✓ Covered
KMT2D	chr12:49,024,558–49,024,728	171	14,000×	✓ Covered
KMT2D	chr12:49,024,790–49,024,966	177	354×	✓ Covered
KMT2D	chr12:49,026,162–49,027,342	1,181	19,848×	✓ Covered
KMT2D	chr12:49,027,783–49,027,950	168	10,031×	✓ Covered
KMT2D	chr12:49,027,989–49,028,161	173	3,436×	✓ Covered
KMT2D	chr12:49,028,808–49,028,978	171	449×	✓ Covered
KMT2D	chr12:49,029,041–49,029,256	216	205×	✓ Covered
KMT2D	chr12:49,029,381–49,029,496	116	1,338×	✓ Covered
KMT2D	chr12:49,030,260–49,030,459	200	350×	✓ Covered
KMT2D	chr12:49,030,581–49,030,788	208	2,714×	✓ Covered
KMT2D	chr12:49,030,873–49,031,053	181	1,573×	✓ Covered
KMT2D	chr12:49,031,155–49,033,984	2,830	2,782×	✓ Covered
KMT2D	chr12:49,034,047–49,034,319	273	2,822×	✓ Covered
KMT2D	chr12:49,034,390–49,034,496	107	508×	✓ Covered
KMT2D	chr12:49,034,562–49,034,686	125	4,859×	✓ Covered
KMT2D	chr12:49,034,792–49,034,955	164	1,070×	✓ Covered
KMT2D	chr12:49,037,105–49,039,009	1,905	6,745×	✓ Covered
KMT2D	chr12:49,039,202–49,039,378	177	5,332×	✓ Covered
KMT2D	chr12:49,039,415–49,039,637	223	3,242×	✓ Covered

FULL PANEL COVERAGE REPORT — Hemingway, Ernest M. (NGS-2026-052101)

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KMT2D	chr12:49,039,704–49,041,555	1,852	552×	✓ Covered
KMT2D	chr12:49,041,635–49,041,725	91	18,709×	✓ Covered
KMT2D	chr12:49,041,897–49,042,010	114	386×	✓ Covered
KMT2D	chr12:49,042,069–49,042,350	282	2,483×	✓ Covered
KMT2D	chr12:49,042,541–49,042,665	125	1,565×	✓ Covered
KMT2D	chr12:49,042,721–49,042,898	178	577×	✓ Covered
KMT2D	chr12:49,043,056–49,043,206	151	4,321×	✓ Covered
KMT2D	chr12:49,043,343–49,043,448	106	7,092×	✓ Covered
KMT2D	chr12:49,043,615–49,043,802	188	1,402×	✓ Covered
KMT2D	chr12:49,043,848–49,044,018	171	9,899×	✓ Covered
KMT2D	chr12:49,044,180–49,044,324	145	6,437×	✓ Covered
KMT2D	chr12:49,044,383–49,044,542	160	3,181×	✓ Covered
KMT2D	chr12:49,044,724–49,044,985	262	1,156×	✓ Covered
KMT2D	chr12:49,045,900–49,045,987	88	806×	✓ Covered
KMT2D	chr12:49,046,045–49,046,194	150	2,815×	✓ Covered
KMT2D	chr12:49,046,240–49,046,444	205	4,703×	✓ Covered
KMT2D	chr12:49,046,589–49,046,810	222	2,725×	✓ Covered
KMT2D	chr12:49,047,945–49,048,089	145	620×	✓ Covered
KMT2D	chr12:49,048,639–49,048,789	151	705×	✓ Covered
KMT2D	chr12:49,049,085–49,049,238	154	2,585×	✓ Covered
KMT2D	chr12:49,049,662–49,050,810	1,149	4,519×	✓ Covered
KMT2D	chr12:49,050,866–49,052,444	1,579	5,752×	✓ Covered
KMT2D	chr12:49,052,544–49,052,729	186	2,941×	✓ Covered
KMT2D	chr12:49,052,895–49,053,092	198	456×	✓ Covered
KMT2D	chr12:49,053,187–49,053,341	155	4,668×	✓ Covered
KMT2D	chr12:49,053,456–49,053,661	206	353×	✓ Covered
KMT2D	chr12:49,053,958–49,054,160	203	1,500×	✓ Covered
KMT2D	chr12:49,054,287–49,054,436	150	140×	✓ Covered
KMT2D	chr12:49,054,508–49,054,771	264	929×	✓ Covered
KMT2D	chr12:49,054,880–49,055,046	167	14,943×	✓ Covered
KMT2D	chr12:49,055,256–49,055,344	89	2,458×	✓ Covered
AJUBA				
AJUBA	chr14:22,973,426–22,973,588	163	1,578×	✓ Covered
AJUBA	chr14:22,974,027–22,974,135	109	1,753×	✓ Covered
AJUBA	chr14:22,974,819–22,974,910	92	4,585×	✓ Covered
AJUBA	chr14:22,974,954–22,975,124	171	1,803×	✓ Covered
AJUBA	chr14:22,976,436–22,976,538	103	2,184×	✓ Covered
AJUBA	chr14:22,976,625–22,976,732	108	1,450×	✓ Covered
AJUBA	chr14:22,978,324–22,978,465	142	18,592×	✓ Covered
AJUBA	chr14:22,981,241–22,982,286	1,046	1,366×	✓ Covered
TP53				
TP53	chr17:7,669,592–7,669,710	119	3,667×	✓ Covered
TP53	chr17:7,670,589–7,670,735	147	1,072×	✓ Covered
TP53	chr17:7,673,515–7,673,628	114	2,067×	✓ Covered
TP53	chr17:7,673,681–7,673,857	177	306×	✓ Covered
TP53	chr17:7,674,161–7,674,310	150	5,821×	✓ Covered
TP53	chr17:7,674,839–7,674,991	153	1,257×	✓ Covered
TP53	chr17:7,675,033–7,675,256	224	865×	✓ Covered
TP53	chr17:7,675,974–7,676,292	319	13,520×	✓ Covered
TP53	chr17:7,676,362–7,676,423	62	1,723×	✓ Covered
TP53	chr17:7,676,501–7,676,614	114	2,736×	✓ Covered

FULL PANEL COVERAGE REPORT — Hemingway, Ernest M. (NGS-2026-052101)

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KEAP1				
KEAP1	chr19:10,486,635–10,486,838	204	507×	✓ Covered
KEAP1	chr19:10,489,172–10,489,388	217	16,094×	✓ Covered
KEAP1	chr19:10,489,628–10,489,873	246	8,292×	✓ Covered
KEAP1	chr19:10,491,557–10,492,282	726	16,091×	✓ Covered
KEAP1	chr19:10,499,375–10,500,053	679	1,781×	✓ Covered
NFE2L2				
NFE2L2	chr2:177,233,985–177,234,291	307	2,307×	✓ Covered
CASP8				
CASP8	chr2:201,266,467–201,266,811	345	1,862×	✓ Covered
CASP8	chr2:201,271,496–201,271,641	146	693×	✓ Covered
CASP8	chr2:201,272,618–201,272,796	179	1,011×	✓ Covered
CASP8	chr2:201,272,878–201,272,962	85	777×	✓ Covered
CASP8	chr2:201,274,869–201,274,973	105	549×	✓ Covered
CASP8	chr2:201,276,807–201,276,988	182	2,491×	✓ Covered
CASP8	chr2:201,284,796–201,285,337	542	610×	✓ Covered
CASP8	chr2:201,286,439–201,286,611	173	1,386×	✓ Covered
CUL3				
CUL3	chr2:224,474,228–224,474,396	169	9,737×	✓ Covered
CUL3	chr2:224,478,180–224,478,365	186	614×	✓ Covered
CUL3	chr2:224,481,872–224,482,098	227	600×	✓ Covered
CUL3	chr2:224,495,812–224,495,986	175	1,790×	✓ Covered
CUL3	chr2:224,497,733–224,497,869	137	9,879×	✓ Covered
CUL3	chr2:224,500,343–224,500,507	165	4,309×	✓ Covered
CUL3	chr2:224,502,945–224,503,092	148	1,109×	✓ Covered
CUL3	chr2:224,503,632–224,503,842	211	2,435×	✓ Covered
CUL3	chr2:224,505,936–224,506,152	217	171×	✓ Covered
CUL3	chr2:224,506,838–224,507,023	186	569×	✓ Covered
CUL3	chr2:224,511,334–224,511,602	269	2,868×	✓ Covered
CUL3	chr2:224,513,504–224,513,658	155	7,318×	✓ Covered
CUL3	chr2:224,514,592–224,514,792	201	396×	✓ Covered
CUL3	chr2:224,535,508–224,535,661	154	4,347×	✓ Covered
CUL3	chr2:224,557,639–224,557,876	238	1,496×	✓ Covered
CUL3	chr2:224,584,924–224,585,029	106	348×	✓ Covered
EP300				
EP300	chr22:41,092,985–41,093,118	134	4,472×	✓ Covered
EP300	chr22:41,117,167–41,117,841	675	3,210×	✓ Covered
EP300	chr22:41,125,844–41,126,060	217	4,223×	✓ Covered
EP300	chr22:41,127,467–41,127,768	302	1,087×	✓ Covered
EP300	chr22:41,129,870–41,130,023	154	6,577×	✓ Covered
EP300	chr22:41,131,368–41,131,653	286	16,690×	✓ Covered
EP300	chr22:41,135,793–41,135,926	134	5,618×	✓ Covered
EP300	chr22:41,137,633–41,137,810	178	1,951×	✓ Covered
EP300	chr22:41,140,120–41,140,277	158	2,597×	✓ Covered
EP300	chr22:41,141,028–41,141,242	215	1,048×	✓ Covered
EP300	chr22:41,146,719–41,146,836	118	1,704×	✓ Covered
EP300	chr22:41,147,817–41,147,966	150	4,114×	✓ Covered
EP300	chr22:41,149,018–41,149,195	178	1,604×	✓ Covered
EP300	chr22:41,149,741–41,150,218	478	880×	✓ Covered
EP300	chr22:41,151,813–41,152,032	220	4,067×	✓ Covered
EP300	chr22:41,152,186–41,152,370	185	1,357×	✓ Covered

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Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
EP300	chr22:41,154,975–41,155,133	159	1,575×	✓ Covered
EP300	chr22:41,157,149–41,157,428	280	4,557×	✓ Covered
EP300	chr22:41,158,392–41,158,520	129	21,838×	✓ Covered
EP300	chr22:41,160,622–41,160,742	121	2,343×	✓ Covered
EP300	chr22:41,162,703–41,162,799	97	661×	✓ Covered
EP300	chr22:41,164,033–41,164,150	118	2,259×	✓ Covered
EP300	chr22:41,166,579–41,166,686	108	1,108×	✓ Covered
EP300	chr22:41,168,429–41,168,619	191	2,170×	✓ Covered
EP300	chr22:41,168,701–41,168,887	187	816×	✓ Covered
EP300	chr22:41,169,483–41,169,636	154	1,050×	✓ Covered
EP300	chr22:41,170,386–41,170,591	206	651×	✓ Covered
EP300	chr22:41,172,479–41,172,683	205	1,933×	✓ Covered
EP300	chr22:41,173,603–41,173,804	202	2,343×	✓ Covered
EP300	chr22:41,176,227–41,176,548	322	627×	✓ Covered
EP300	chr22:41,176,753–41,178,973	2,221	3,110×	✓ Covered
PIK3CA				
PIK3CA	chr3:179,218,190–179,218,354	165	3,221×	✓ Covered
PIK3CA	chr3:179,234,074–179,234,381	308	2,449×	● **VARIANT DETECTED**
FBXW7				
FBXW7	chr4:152,325,986–152,326,251	266	4,475×	✓ Covered
FAT1				
FAT1	chr4:186,588,575–186,589,240	666	945×	✓ Covered
FAT1	chr4:186,595,669–186,595,846	178	128×	✓ Covered
FAT1	chr4:186,596,520–186,597,191	672	2,030×	✓ Covered
FAT1	chr4:186,597,662–186,597,812	151	235×	✓ Covered
FAT1	chr4:186,597,952–186,598,145	194	1,870×	✓ Covered
FAT1	chr4:186,599,878–186,600,380	503	2,398×	✓ Covered
FAT1	chr4:186,601,249–186,601,446	198	1,210×	✓ Covered
FAT1	chr4:186,602,883–186,603,054	172	616×	✓ Covered
FAT1	chr4:186,603,156–186,603,997	842	582×	✓ Covered
FAT1	chr4:186,604,357–186,604,594	238	645×	✓ Covered
FAT1	chr4:186,606,050–186,606,233	184	2,133×	✓ Covered
FAT1	chr4:186,609,163–186,609,340	178	4,000×	✓ Covered
FAT1	chr4:186,609,781–186,610,035	255	1,708×	✓ Covered
FAT1	chr4:186,611,366–186,611,795	430	3,935×	✓ Covered
FAT1	chr4:186,613,089–186,613,362	274	6,152×	✓ Covered
FAT1	chr4:186,614,171–186,614,364	194	2,390×	✓ Covered
FAT1	chr4:186,616,985–186,617,221	237	7,970×	✓ Covered
FAT1	chr4:186,617,688–186,621,795	4,108	3,260×	✓ Covered
FAT1	chr4:186,628,134–186,628,384	251	2,847×	✓ Covered
FAT1	chr4:186,628,468–186,628,783	316	11,410×	✓ Covered
FAT1	chr4:186,633,664–186,633,843	180	1,988×	✓ Covered
FAT1	chr4:186,636,005–186,636,255	251	2,823×	✓ Covered
FAT1	chr4:186,636,565–186,636,934	370	8,535×	✓ Covered
FAT1	chr4:186,639,702–186,639,803	102	1,240×	✓ Covered
FAT1	chr4:186,663,279–186,663,633	355	5,334×	✓ Covered
FAT1	chr4:186,706,543–186,709,847	3,305	2,289×	✓ Covered
TERT				
TERT	chr5:1,295,050–1,295,250	201	3,866×	✓ Covered
PIK3R1				
PIK3R1	chr5:68,294,516–68,294,698	183	7,278×	✓ Covered

FULL PANEL COVERAGE REPORT — Hemingway, Ernest M. (NGS-2026-052101)

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
PIK3R1	chr5:68,295,128–68,295,344	217	958×	✓ Covered
PIK3R1	chr5:68,295,400–68,295,508	109	7,995×	✓ Covered
NSD1				
NSD1	chr5:177,135,084–177,136,050	967	2,152×	✓ Covered
NSD1	chr5:177,191,864–177,192,039	176	6,751×	✓ Covered
NSD1	chr5:177,204,100–177,204,312	213	331×	✓ Covered
NSD1	chr5:177,209,616–177,212,215	2,600	2,391×	✓ Covered
NSD1	chr5:177,235,801–177,235,965	165	18,625×	✓ Covered
NSD1	chr5:177,238,217–177,238,527	311	587×	✓ Covered
NSD1	chr5:177,239,736–177,239,885	150	2,040×	✓ Covered
NSD1	chr5:177,244,175–177,244,290	116	2,120×	✓ Covered
NSD1	chr5:177,246,658–177,246,816	159	11,444×	✓ Covered
NSD1	chr5:177,248,161–177,248,344	184	679×	✓ Covered
NSD1	chr5:177,251,710–177,251,873	164	3,162×	✓ Covered
NSD1	chr5:177,256,931–177,257,171	241	3,319×	✓ Covered
NSD1	chr5:177,259,969–177,260,188	220	3,015×	✓ Covered
NSD1	chr5:177,267,542–177,267,738	197	1,664×	✓ Covered
NSD1	chr5:177,269,582–177,269,827	246	411×	✓ Covered
NSD1	chr5:177,273,652–177,273,804	153	1,473×	✓ Covered
NSD1	chr5:177,280,545–177,280,854	310	11,901×	✓ Covered
NSD1	chr5:177,282,445–177,282,601	157	5,923×	✓ Covered
NSD1	chr5:177,283,767–177,283,948	182	8,604×	✓ Covered
NSD1	chr5:177,288,799–177,288,945	147	2,557×	✓ Covered
NSD1	chr5:177,291,934–177,292,178	245	6,936×	✓ Covered
NSD1	chr5:177,293,812–177,295,476	1,665	973×	✓ Covered
EGFR				
EGFR	chr7:55,173,901–55,174,063	163	451×	✓ Covered
EGFR	chr7:55,174,702–55,174,840	139	4,279×	✓ Covered
EGFR	chr7:55,181,273–55,181,498	226	1,233×	✓ Covered
EGFR	chr7:55,191,699–55,191,894	196	1,872×	✓ Covered
CDKN2A				
CDKN2A	chr9:21,968,212–21,968,262	51	764×	✓ Covered
CDKN2A	chr9:21,970,882–21,971,228	347	727×	✓ Covered
CDKN2A	chr9:21,974,658–21,974,846	189	2,416×	✓ Covered
CDKN2A	chr9:21,974,658–21,974,847	190	1,547×	✓ Covered
NOTCH1				
NOTCH1	chr9:136,496,054–136,497,578	1,525	1,803×	✓ Covered
NOTCH1	chr9:136,498,879–136,499,016	138	609×	✓ Covered
NOTCH1	chr9:136,499,092–136,499,279	188	1,013×	✓ Covered
NOTCH1	chr9:136,500,532–136,500,867	336	2,006×	✓ Covered
NOTCH1	chr9:136,501,728–136,501,933	206	537×	✓ Covered
NOTCH1	chr9:136,501,981–136,502,108	128	431×	✓ Covered
NOTCH1	chr9:136,502,252–136,502,508	257	5,612×	✓ Covered
NOTCH1	chr9:136,503,162–136,503,350	189	5,097×	✓ Covered
NOTCH1	chr9:136,504,653–136,505,124	472	7,438×	✓ Covered
NOTCH1	chr9:136,505,290–136,505,901	612	26,740×	✓ Covered
NOTCH1	chr9:136,506,507–136,506,659	153	1,973×	✓ Covered
NOTCH1	chr9:136,506,696–136,506,993	298	3,678×	✓ Covered
NOTCH1	chr9:136,507,285–136,507,457	173	4,495×	✓ Covered
NOTCH1	chr9:136,507,935–136,508,159	225	2,914×	✓ Covered
NOTCH1	chr9:136,508,212–136,508,405	194	2,547×	✓ Covered

FULL PANEL COVERAGE REPORT — Hemingway, Ernest M. (NGS-2026-052101)

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
NOTCH1	chr9:136,508,850–136,509,091	242	336×	✓ Covered
NOTCH1	chr9:136,509,713–136,509,981	269	12,265×	✓ Covered
NOTCH1	chr9:136,510,633–136,510,825	193	499×	✓ Covered
NOTCH1	chr9:136,511,132–136,511,291	160	763×	✓ Covered
NOTCH1	chr9:136,513,001–136,513,154	154	4,546×	✓ Covered
NOTCH1	chr9:136,513,372–136,513,557	186	226×	✓ Covered
NOTCH1	chr9:136,514,490–136,514,722	233	1,114×	✓ Covered
NOTCH1	chr9:136,515,270–136,515,420	151	1,115×	✓ Covered
NOTCH1	chr9:136,515,463–136,515,736	274	76,555×	✓ Covered
NOTCH1	chr9:136,515,961–136,516,114	154	2,781×	✓ Covered
NOTCH1	chr9:136,517,252–136,517,405	154	461×	✓ Covered
NOTCH1	chr9:136,517,732–136,517,957	226	2,098×	✓ Covered
NOTCH1	chr9:136,518,117–136,518,312	196	2,064×	✓ Covered
NOTCH1	chr9:136,518,571–136,518,844	274	6,167×	✓ Covered
NOTCH1	chr9:136,519,423–136,519,585	163	7,295×	✓ Covered
NOTCH1	chr9:136,522,830–136,523,208	379	1,091×	✓ Covered
NOTCH1	chr9:136,523,697–136,523,999	303	137×	✓ Covered
NOTCH1	chr9:136,544,004–136,544,122	119	1,142×	✓ Covered
NOTCH1	chr9:136,545,706–136,545,806	101	12,250×	✓ Covered

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PATIENT & SPECIMEN INFORMATION

Patient name	Woolf, Virginia A.	**Accession**	NGS-2026-052102
Date of birth	1970-03-11 (Age 56)	**Specimen**	Oral swab, right buccal mucosa
MRN	20456732	**Collected**	2026-05-19
Ordering physician	Dr. Wisdom, MD	**Received**	2026-05-19
Clinical indication	Oral lesion — right buccal mucosa	**Reported**	2026-05-20

RESULT: VARIANTS DETECTED

DETECTED VARIANTS

Gene	HGVS (coding)	Protein change	VAF	Depth	Tier	Classification
NOTCH1	NM_017617.5:c.7244C>T	p.Pro2415Leu	0.26%	7,690×	II	Likely Pathogenic

VARIANT ANNOTATION

NOTCH1 p.Pro2415Leu | Likely Pathogenic (Tier II)

NOTCH1 functions as a tumour suppressor in stratified squamous epithelium. The p.Pro2415Leu missense variant affects a conserved residue in the PEST domain, predicted to disrupt protein stability (SIFT: deleterious; PolyPhen-2: probably damaging). NOTCH1 loss-of-function mutations are found in 10–15% of OSCC and are associated with early-onset tumourigenesis in the oral cavity. This variant is classified Likely Pathogenic (Tier II). In the context of confirmed dysplasia, detection of a NOTCH1 tumour suppressor variant supports malignant progression risk. Clinical correlation and multidisciplinary review are recommended. NOTE: Observed VAF (0.26%) is above the 0.2% reporting threshold and is supported by 2 consensus reads at 769× consensus depth. Orthogonal confirmation is recommended before clinical action.

GENES WITH NO PATHOGENIC OR LIKELY PATHOGENIC VARIANTS DETECTED

AJUBA, CASP8, CDKN2A, CUL3, DPYD, EGFR, EP300, FAT1, FBXW7, HRAS, KEAP1, KMT2D, KRAS, NFE2L2, NSD1, PIK3CA, PIK3R1, PTEN, TERT, TP53

Variants of uncertain significance (Tier III) are not reported.

METHOD SUMMARY

Cell-free and cellular DNA extracted from oral swab is subjected to hybridization capture using the IDT xGen Duplex UMI panel targeting 251 genomic regions (~77 kbp) across 21 cancer-relevant genes. Libraries incorporate 9-bp Unique Molecular Identifiers (UMIs) in a 9M1S+T read structure on both R1 and R2. Libraries are sequenced on Illumina NextSeq (2×150 bp). After alignment to GRCh38 (canonical chromosomes), reads are grouped by UMI using fgbio (adjacency strategy, ≤1 edit distance), and molecular consensus sequences are called requiring ≥2 reads per UMI family. Somatic variants are called on the consensus BAM with VarDict at minimum allele frequency 0.001. Variants ≥0.2% consensus VAF with ≥2 consensus supporting reads are reported. UMI deduplication reduces sequencing noise by >800-fold. Run-level negative controls are included on every sequencing run.

LIMITATIONS

This assay detects somatic single nucleotide variants and small indels within 251 target regions (~77 kbp) at VAF ≥0.2% with validated sensitivity of 98.4% at ≥20,000× raw input depth (~4,000× consensus after UMI deduplication). Structural variants, copy number alterations, gene fusions, large indels, intronic variants outside defined capture regions, and variants in regions with low hybridization capture efficiency are not reliably assessed. A negative result does not exclude malignancy. Tumour heterogeneity, low tumour cell content, and specimen quality may affect variant allele fraction. Variants below the 0.2% VAF reporting threshold may be present but are not reported. At 20,000× raw sequencing depth (~4,000× mean consensus), approximately 0.3% of targets may fall below 1,000× consensus depth; per-target coverage is provided in the laboratory technical report. Results should be interpreted in the context of clinical and pathological findings by a qualified clinician.

Reported by: Dr. Wisdom, MD PhD, FCAP **Report date: 2026-05-20**

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

HOTSPOT SCREENING SUMMARY

Patient: Woolf, Virginia A. **Accession:** NGS-2026-052102

Hotspots listed below were explicitly interrogated in addition to all variants above 0.2 % VAF with ≥2 consensus reads across all 251 targets. Coverage criteria (≥100× consensus depth) were met at all targets.

EGFR		Cov ✓
L858R	c.2573T>G	Not detected
T790M	c.2369C>T	Not detected
Exon 19 del	c.2235_2249del	Not detected

KRAS		Cov ✓
G12D	c.35G>T	Not detected
G12V	c.35G>A	Not detected
G12C	c.34G>T	Not detected
G13D	c.38G>A	Not detected

PIK3CA		Cov ✓
E542K	c.1624G>A	Not detected
E545K	c.1633G>A	Not detected
H1047R	c.3140A>G	Not detected

TERT		Cov ✓
C228T	c.-124C>T	Not detected
C250T	c.-146C>T	Not detected

HRAS		Cov ✓
G12V	c.35G>T	Not detected
G12D	c.35G>A	Not detected
Q61L	c.182A>T	Not detected
Q61R	c.182A>G	Not detected

NOTCH1		Cov ✓
P2415L	c.7244C>T	● **DETECTED**
R1699W	c.5095C>T	Not detected

TP53		Cov ✓
R175H	c.524G>A	Not detected
R248Q	c.743G>A	Not detected
R248W	c.742C>T	Not detected
R273H	c.818G>A	Not detected
R282W	c.844C>T	Not detected

REMAINING PANEL GENES — SEQUENCE-LEVEL DETECTION, NO PREDEFINED HOTSPOT LIST

AJUBA · CASP8 · CDKN2A · CUL3 · DPYD · EP300 · FAT1 · FBXW7 · KEAP1 · KMT2D · NFE2L2 · NSD1 · PIK3R1 · PTEN

All variants above 0.2 % VAF with ≥2 consensus reads are reported for these genes. No variants meeting reporting criteria were detected. All targets in these genes met minimum coverage criteria (≥100× consensus depth).

Reported by: Dr. Wisdom, MD PhD, FCAP **Report date: 2026-05-20**

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

FULL PANEL COVERAGE REPORT — Woolf, Virginia A. (NGS-2026-052102)

251 targets, 77,417 bp total. Mean consensus depth: 4,438×. Targets below 100× consensus: 0 / 251 (0%). Depths are derived from the consensus BAM (UMI deduplication, min-reads ≥ 2). Positions reported 1-based.

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
DPYD				
DPYD	chr1:97,082,291–97,082,491	201	1,840×	✓ Covered
DPYD	chr1:97,449,958–97,450,158	201	1,977×	✓ Covered
DPYD	chr1:97,515,687–97,515,887	201	3,397×	✓ Covered
DPYD	chr1:97,573,763–97,573,963	201	2,077×	✓ Covered
PTEN				
PTEN	chr10:87,864,450–87,864,568	119	1,628×	✓ Covered
PTEN	chr10:87,894,005–87,894,129	125	4,197×	✓ Covered
PTEN	chr10:87,925,493–87,925,577	85	2,185×	✓ Covered
PTEN	chr10:87,931,026–87,931,109	84	1,735×	✓ Covered
PTEN	chr10:87,932,993–87,933,271	279	5,451×	✓ Covered
PTEN	chr10:87,952,098–87,952,279	182	1,771×	✓ Covered
PTEN	chr10:87,957,833–87,958,039	207	541×	✓ Covered
PTEN	chr10:87,960,874–87,961,138	265	4,987×	✓ Covered
PTEN	chr10:87,965,267–87,965,489	223	1,999×	✓ Covered
HRAS				
HRAS	chr11:532,619–532,775	157	5,465×	✓ Covered
HRAS	chr11:533,433–533,632	200	1,719×	✓ Covered
HRAS	chr11:533,746–533,964	219	238×	✓ Covered
HRAS	chr11:534,192–534,342	151	557×	✓ Covered
KRAS				
KRAS	chr12:25,209,778–25,209,931	154	991×	✓ Covered
KRAS	chr12:25,225,594–25,225,793	200	451×	✓ Covered
KRAS	chr12:25,227,214–25,227,432	219	890×	✓ Covered
KRAS	chr12:25,245,254–25,245,404	151	5,721×	✓ Covered
KMT2D				
KMT2D	chr12:49,021,763–49,021,892	130	577×	✓ Covered
KMT2D	chr12:49,022,023–49,022,171	149	31,226×	✓ Covered
KMT2D	chr12:49,022,260–49,022,373	114	566×	✓ Covered
KMT2D	chr12:49,022,570–49,022,895	326	4,597×	✓ Covered
KMT2D	chr12:49,024,558–49,024,728	171	1,749×	✓ Covered
KMT2D	chr12:49,024,790–49,024,966	177	226×	✓ Covered
KMT2D	chr12:49,026,162–49,027,342	1,181	2,075×	✓ Covered
KMT2D	chr12:49,027,783–49,027,950	168	2,420×	✓ Covered
KMT2D	chr12:49,027,989–49,028,161	173	2,934×	✓ Covered
KMT2D	chr12:49,028,808–49,028,978	171	1,306×	✓ Covered
KMT2D	chr12:49,029,041–49,029,256	216	184×	✓ Covered
KMT2D	chr12:49,029,381–49,029,496	116	2,045×	✓ Covered
KMT2D	chr12:49,030,260–49,030,459	200	1,222×	✓ Covered
KMT2D	chr12:49,030,581–49,030,788	208	1,893×	✓ Covered
KMT2D	chr12:49,030,873–49,031,053	181	763×	✓ Covered
KMT2D	chr12:49,031,155–49,033,984	2,830	1,493×	✓ Covered
KMT2D	chr12:49,034,047–49,034,319	273	14,910×	✓ Covered
KMT2D	chr12:49,034,390–49,034,496	107	1,732×	✓ Covered
KMT2D	chr12:49,034,562–49,034,686	125	648×	✓ Covered
KMT2D	chr12:49,034,792–49,034,955	164	755×	✓ Covered
KMT2D	chr12:49,037,105–49,039,009	1,905	3,846×	✓ Covered
KMT2D	chr12:49,039,202–49,039,378	177	5,756×	✓ Covered
KMT2D	chr12:49,039,415–49,039,637	223	19,638×	✓ Covered

FULL PANEL COVERAGE REPORT — Woolf, Virginia A. (NGS-2026-052102)

Gene	chr:start–end (1-based)	Len (bp)	Depth	Coverage status
KMT2D	chr12:49,039,704–49,041,555	1,852	5,682×	✓ Covered
KMT2D	chr12:49,041,635–49,041,725	91	778×	✓ Covered
KMT2D	chr12:49,041,897–49,042,010	114	9,454×	✓ Covered
KMT2D	chr12:49,042,069–49,042,350	282	14,703×	✓ Covered
KMT2D	chr12:49,042,541–49,042,665	125	2,797×	✓ Covered
KMT2D	chr12:49,042,721–49,042,898	178	2,832×	✓ Covered
KMT2D	chr12:49,043,056–49,043,206	151	3,253×	✓ Covered
KMT2D	chr12:49,043,343–49,043,448	106	12,590×	✓ Covered
KMT2D	chr12:49,043,615–49,043,802	188	2,471×	✓ Covered
KMT2D	chr12:49,043,848–49,044,018	171	4,045×	✓ Covered
KMT2D	chr12:49,044,180–49,044,324	145	600×	✓ Covered
KMT2D	chr12:49,044,383–49,044,542	160	643×	✓ Covered
KMT2D	chr12:49,044,724–49,044,985	262	2,100×	✓ Covered
KMT2D	chr12:49,045,900–49,045,987	88	6,413×	✓ Covered
KMT2D	chr12:49,046,045–49,046,194	150	4,195×	✓ Covered
KMT2D	chr12:49,046,240–49,046,444	205	1,359×	✓ Covered
KMT2D	chr12:49,046,589–49,046,810	222	2,826×	✓ Covered
KMT2D	chr12:49,047,945–49,048,089	145	16,274×	✓ Covered
KMT2D	chr12:49,048,639–49,048,789	151	34,669×	✓ Covered
KMT2D	chr12:49,049,085–49,049,238	154	1,238×	✓ Covered
KMT2D	chr12:49,049,662–49,050,810	1,149	8,585×	✓ Covered
KMT2D	chr12:49,050,866–49,052,444	1,579	9,541×	✓ Covered
KMT2D	chr12:49,052,544–49,052,729	186	2,274×	✓ Covered
KMT2D	chr12:49,052,895–49,053,092	198	7,055×	✓ Covered
KMT2D	chr12:49,053,187–49,053,341	155	1,325×	✓ Covered
KMT2D	chr12:49,053,456–49,053,661	206	5,134×	✓ Covered
KMT2D	chr12:49,053,958–49,054,160	203	1,309×	✓ Covered
KMT2D	chr12:49,054,287–49,054,436	150	393×	✓ Covered
KMT2D	chr12:49,054,508–49,054,771	264	2,383×	✓ Covered
KMT2D	chr12:49,054,880–49,055,046	167	11,184×	✓ Covered
KMT2D	chr12:49,055,256–49,055,344	89	3,075×	✓ Covered
AJUBA				
AJUBA	chr14:22,973,426–22,973,588	163	848×	✓ Covered
AJUBA	chr14:22,974,027–22,974,135	109	341×	✓ Covered
AJUBA	chr14:22,974,819–22,974,910	92	1,209×	✓ Covered
AJUBA	chr14:22,974,954–22,975,124	171	981×	✓ Covered
AJUBA	chr14:22,976,436–22,976,538	103	681×	✓ Covered
AJUBA	chr14:22,976,625–22,976,732	108	2,742×	✓ Covered
AJUBA	chr14:22,978,324–22,978,465	142	802×	✓ Covered
AJUBA	chr14:22,981,241–22,982,286	1,046	2,078×	✓ Covered
TP53				
TP53	chr17:7,669,592–7,669,710	119	2,118×	✓ Covered
TP53	chr17:7,670,589–7,670,735	147	4,452×	✓ Covered
TP53	chr17:7,673,515–7,673,628	114	255×	✓ Covered
TP53	chr17:7,673,681–7,673,857	177	377×	✓ Covered
TP53	chr17:7,674,161–7,674,310	150	2,703×	✓ Covered
TP53	chr17:7,674,839–7,674,991	153	4,212×	✓ Covered
TP53	chr17:7,675,033–7,675,256	224	7,746×	✓ Covered
TP53	chr17:7,675,974–7,676,292	319	2,810×	✓ Covered
TP53	chr17:7,676,362–7,676,423	62	422×	✓ Covered
TP53	chr17:7,676,501–7,676,614	114	115×	✓ Covered

FULL PANEL COVERAGE REPORT — Woolf, Virginia A. (NGS-2026-052102)

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KEAP1				
KEAP1	chr19:10,486,635–10,486,838	204	964×	✓ Covered
KEAP1	chr19:10,489,172–10,489,388	217	901×	✓ Covered
KEAP1	chr19:10,489,628–10,489,873	246	540×	✓ Covered
KEAP1	chr19:10,491,557–10,492,282	726	949×	✓ Covered
KEAP1	chr19:10,499,375–10,500,053	679	25,254×	✓ Covered
NFE2L2				
NFE2L2	chr2:177,233,985–177,234,291	307	2,737×	✓ Covered
CASP8				
CASP8	chr2:201,266,467–201,266,811	345	263×	✓ Covered
CASP8	chr2:201,271,496–201,271,641	146	11,154×	✓ Covered
CASP8	chr2:201,272,618–201,272,796	179	1,019×	✓ Covered
CASP8	chr2:201,272,878–201,272,962	85	2,500×	✓ Covered
CASP8	chr2:201,274,869–201,274,973	105	5,804×	✓ Covered
CASP8	chr2:201,276,807–201,276,988	182	1,757×	✓ Covered
CASP8	chr2:201,284,796–201,285,337	542	8,929×	✓ Covered
CASP8	chr2:201,286,439–201,286,611	173	1,099×	✓ Covered
CUL3				
CUL3	chr2:224,474,228–224,474,396	169	1,364×	✓ Covered
CUL3	chr2:224,478,180–224,478,365	186	6,986×	✓ Covered
CUL3	chr2:224,481,872–224,482,098	227	296×	✓ Covered
CUL3	chr2:224,495,812–224,495,986	175	197×	✓ Covered
CUL3	chr2:224,497,733–224,497,869	137	15,968×	✓ Covered
CUL3	chr2:224,500,343–224,500,507	165	255×	✓ Covered
CUL3	chr2:224,502,945–224,503,092	148	893×	✓ Covered
CUL3	chr2:224,503,632–224,503,842	211	1,263×	✓ Covered
CUL3	chr2:224,505,936–224,506,152	217	1,241×	✓ Covered
CUL3	chr2:224,506,838–224,507,023	186	7,878×	✓ Covered
CUL3	chr2:224,511,334–224,511,602	269	424×	✓ Covered
CUL3	chr2:224,513,504–224,513,658	155	2,225×	✓ Covered
CUL3	chr2:224,514,592–224,514,792	201	3,026×	✓ Covered
CUL3	chr2:224,535,508–224,535,661	154	1,601×	✓ Covered
CUL3	chr2:224,557,639–224,557,876	238	1,010×	✓ Covered
CUL3	chr2:224,584,924–224,585,029	106	1,348×	✓ Covered
EP300				
EP300	chr22:41,092,985–41,093,118	134	504×	✓ Covered
EP300	chr22:41,117,167–41,117,841	675	445×	✓ Covered
EP300	chr22:41,125,844–41,126,060	217	3,033×	✓ Covered
EP300	chr22:41,127,467–41,127,768	302	2,169×	✓ Covered
EP300	chr22:41,129,870–41,130,023	154	2,304×	✓ Covered
EP300	chr22:41,131,368–41,131,653	286	19,102×	✓ Covered
EP300	chr22:41,135,793–41,135,926	134	3,221×	✓ Covered
EP300	chr22:41,137,633–41,137,810	178	3,265×	✓ Covered
EP300	chr22:41,140,120–41,140,277	158	2,401×	✓ Covered
EP300	chr22:41,141,028–41,141,242	215	1,895×	✓ Covered
EP300	chr22:41,146,719–41,146,836	118	2,690×	✓ Covered
EP300	chr22:41,147,817–41,147,966	150	577×	✓ Covered
EP300	chr22:41,149,018–41,149,195	178	724×	✓ Covered
EP300	chr22:41,149,741–41,150,218	478	608×	✓ Covered
EP300	chr22:41,151,813–41,152,032	220	3,739×	✓ Covered
EP300	chr22:41,152,186–41,152,370	185	961×	✓ Covered

FULL PANEL COVERAGE REPORT — Woolf, Virginia A. (NGS-2026-052102)

Gene	chr:start–end (1-based)	Len (bp)	Depth	Coverage status
EP300	chr22:41,154,975–41,155,133	159	4,345×	✓ Covered
EP300	chr22:41,157,149–41,157,428	280	780×	✓ Covered
EP300	chr22:41,158,392–41,158,520	129	447×	✓ Covered
EP300	chr22:41,160,622–41,160,742	121	2,887×	✓ Covered
EP300	chr22:41,162,703–41,162,799	97	538×	✓ Covered
EP300	chr22:41,164,033–41,164,150	118	2,883×	✓ Covered
EP300	chr22:41,166,579–41,166,686	108	4,708×	✓ Covered
EP300	chr22:41,168,429–41,168,619	191	8,726×	✓ Covered
EP300	chr22:41,168,701–41,168,887	187	2,187×	✓ Covered
EP300	chr22:41,169,483–41,169,636	154	18,555×	✓ Covered
EP300	chr22:41,170,386–41,170,591	206	3,479×	✓ Covered
EP300	chr22:41,172,479–41,172,683	205	13,796×	✓ Covered
EP300	chr22:41,173,603–41,173,804	202	998×	✓ Covered
EP300	chr22:41,176,227–41,176,548	322	1,106×	✓ Covered
EP300	chr22:41,176,753–41,178,973	2,221	717×	✓ Covered
PIK3CA				
PIK3CA	chr3:179,218,190–179,218,354	165	637×	✓ Covered
PIK3CA	chr3:179,234,074–179,234,381	308	2,628×	✓ Covered
FBXW7				
FBXW7	chr4:152,325,986–152,326,251	266	4,951×	✓ Covered
FAT1				
FAT1	chr4:186,588,575–186,589,240	666	4,368×	✓ Covered
FAT1	chr4:186,595,669–186,595,846	178	380×	✓ Covered
FAT1	chr4:186,596,520–186,597,191	672	679×	✓ Covered
FAT1	chr4:186,597,662–186,597,812	151	2,080×	✓ Covered
FAT1	chr4:186,597,952–186,598,145	194	540×	✓ Covered
FAT1	chr4:186,599,878–186,600,380	503	471×	✓ Covered
FAT1	chr4:186,601,249–186,601,446	198	718×	✓ Covered
FAT1	chr4:186,602,883–186,603,054	172	1,210×	✓ Covered
FAT1	chr4:186,603,156–186,603,997	842	5,766×	✓ Covered
FAT1	chr4:186,604,357–186,604,594	238	1,616×	✓ Covered
FAT1	chr4:186,606,050–186,606,233	184	2,989×	✓ Covered
FAT1	chr4:186,609,163–186,609,340	178	5,383×	✓ Covered
FAT1	chr4:186,609,781–186,610,035	255	270×	✓ Covered
FAT1	chr4:186,611,366–186,611,795	430	474×	✓ Covered
FAT1	chr4:186,613,089–186,613,362	274	2,470×	✓ Covered
FAT1	chr4:186,614,171–186,614,364	194	1,887×	✓ Covered
FAT1	chr4:186,616,985–186,617,221	237	844×	✓ Covered
FAT1	chr4:186,617,688–186,621,795	4,108	682×	✓ Covered
FAT1	chr4:186,628,134–186,628,384	251	1,064×	✓ Covered
FAT1	chr4:186,628,468–186,628,783	316	1,152×	✓ Covered
FAT1	chr4:186,633,664–186,633,843	180	4,243×	✓ Covered
FAT1	chr4:186,636,005–186,636,255	251	3,171×	✓ Covered
FAT1	chr4:186,636,565–186,636,934	370	1,530×	✓ Covered
FAT1	chr4:186,639,702–186,639,803	102	1,287×	✓ Covered
FAT1	chr4:186,663,279–186,663,633	355	8,717×	✓ Covered
FAT1	chr4:186,706,543–186,709,847	3,305	633×	✓ Covered
TERT				
TERT	chr5:1,295,050–1,295,250	201	4,447×	✓ Covered
PIK3R1				
PIK3R1	chr5:68,294,516–68,294,698	183	1,227×	✓ Covered

FULL PANEL COVERAGE REPORT — Woolf, Virginia A. (NGS-2026-052102)

Gene	chr:start–end (1-based)	Len (bp)	Depth	Coverage status
PIK3R1	chr5:68,295,128–68,295,344	217	874×	✓ Covered
PIK3R1	chr5:68,295,400–68,295,508	109	812×	✓ Covered
NSD1				
NSD1	chr5:177,135,084–177,136,050	967	262×	✓ Covered
NSD1	chr5:177,191,864–177,192,039	176	3,510×	✓ Covered
NSD1	chr5:177,204,100–177,204,312	213	3,934×	✓ Covered
NSD1	chr5:177,209,616–177,212,215	2,600	2,914×	✓ Covered
NSD1	chr5:177,235,801–177,235,965	165	5,274×	✓ Covered
NSD1	chr5:177,238,217–177,238,527	311	179×	✓ Covered
NSD1	chr5:177,239,736–177,239,885	150	6,137×	✓ Covered
NSD1	chr5:177,244,175–177,244,290	116	1,805×	✓ Covered
NSD1	chr5:177,246,658–177,246,816	159	167,496×	✓ Covered
NSD1	chr5:177,248,161–177,248,344	184	363×	✓ Covered
NSD1	chr5:177,251,710–177,251,873	164	57,752×	✓ Covered
NSD1	chr5:177,256,931–177,257,171	241	2,717×	✓ Covered
NSD1	chr5:177,259,969–177,260,188	220	662×	✓ Covered
NSD1	chr5:177,267,542–177,267,738	197	10,950×	✓ Covered
NSD1	chr5:177,269,582–177,269,827	246	2,401×	✓ Covered
NSD1	chr5:177,273,652–177,273,804	153	20,923×	✓ Covered
NSD1	chr5:177,280,545–177,280,854	310	368×	✓ Covered
NSD1	chr5:177,282,445–177,282,601	157	1,426×	✓ Covered
NSD1	chr5:177,283,767–177,283,948	182	2,023×	✓ Covered
NSD1	chr5:177,288,799–177,288,945	147	1,545×	✓ Covered
NSD1	chr5:177,291,934–177,292,178	245	3,580×	✓ Covered
NSD1	chr5:177,293,812–177,295,476	1,665	3,844×	✓ Covered
EGFR				
EGFR	chr7:55,173,901–55,174,063	163	2,261×	✓ Covered
EGFR	chr7:55,174,702–55,174,840	139	6,018×	✓ Covered
EGFR	chr7:55,181,273–55,181,498	226	1,153×	✓ Covered
EGFR	chr7:55,191,699–55,191,894	196	933×	✓ Covered
CDKN2A				
CDKN2A	chr9:21,968,212–21,968,262	51	4,658×	✓ Covered
CDKN2A	chr9:21,970,882–21,971,228	347	1,866×	✓ Covered
CDKN2A	chr9:21,974,658–21,974,846	189	6,507×	✓ Covered
CDKN2A	chr9:21,974,658–21,974,847	190	991×	✓ Covered
NOTCH1				
NOTCH1	chr9:136,496,054–136,497,578	1,525	2,214×	● **VARIANT DETECTED**
NOTCH1	chr9:136,498,879–136,499,016	138	634×	✓ Covered
NOTCH1	chr9:136,499,092–136,499,279	188	7,162×	✓ Covered
NOTCH1	chr9:136,500,532–136,500,867	336	1,115×	✓ Covered
NOTCH1	chr9:136,501,728–136,501,933	206	13,972×	✓ Covered
NOTCH1	chr9:136,501,981–136,502,108	128	1,373×	✓ Covered
NOTCH1	chr9:136,502,252–136,502,508	257	615×	✓ Covered
NOTCH1	chr9:136,503,162–136,503,350	189	1,443×	✓ Covered
NOTCH1	chr9:136,504,653–136,505,124	472	9,800×	✓ Covered
NOTCH1	chr9:136,505,290–136,505,901	612	1,638×	✓ Covered
NOTCH1	chr9:136,506,507–136,506,659	153	1,688×	✓ Covered
NOTCH1	chr9:136,506,696–136,506,993	298	2,286×	✓ Covered
NOTCH1	chr9:136,507,285–136,507,457	173	4,787×	✓ Covered
NOTCH1	chr9:136,507,935–136,508,159	225	1,927×	✓ Covered
NOTCH1	chr9:136,508,212–136,508,405	194	23,004×	✓ Covered

FULL PANEL COVERAGE REPORT — Woolf, Virginia A. (NGS-2026-052102)

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
NOTCH1	chr9:136,508,850–136,509,091	242	653×	✓ Covered
NOTCH1	chr9:136,509,713–136,509,981	269	1,807×	✓ Covered
NOTCH1	chr9:136,510,633–136,510,825	193	1,924×	✓ Covered
NOTCH1	chr9:136,511,132–136,511,291	160	1,868×	✓ Covered
NOTCH1	chr9:136,513,001–136,513,154	154	3,411×	✓ Covered
NOTCH1	chr9:136,513,372–136,513,557	186	771×	✓ Covered
NOTCH1	chr9:136,514,490–136,514,722	233	1,114×	✓ Covered
NOTCH1	chr9:136,515,270–136,515,420	151	1,372×	✓ Covered
NOTCH1	chr9:136,515,463–136,515,736	274	8,274×	✓ Covered
NOTCH1	chr9:136,515,961–136,516,114	154	2,265×	✓ Covered
NOTCH1	chr9:136,517,252–136,517,405	154	293×	✓ Covered
NOTCH1	chr9:136,517,732–136,517,957	226	3,024×	✓ Covered
NOTCH1	chr9:136,518,117–136,518,312	196	2,764×	✓ Covered
NOTCH1	chr9:136,518,571–136,518,844	274	289×	✓ Covered
NOTCH1	chr9:136,519,423–136,519,585	163	453×	✓ Covered
NOTCH1	chr9:136,522,830–136,523,208	379	2,343×	✓ Covered
NOTCH1	chr9:136,523,697–136,523,999	303	3,033×	✓ Covered
NOTCH1	chr9:136,544,004–136,544,122	119	17,761×	✓ Covered
NOTCH1	chr9:136,545,706–136,545,806	101	323×	✓ Covered

PATIENT & SPECIMEN INFORMATION

Patient name	Dickens, Charles J.	**Accession**	NGS-2026-052103
Date of birth	1951-09-28 (Age 74)	**Specimen**	Oral swab, left floor of mouth
MRN	30567843	**Collected**	2026-05-19
Ordering physician	Dr. Wisdom, DDS MD	**Received**	2026-05-19
Clinical indication	Oral lesion — left floor of mouth	**Reported**	2026-05-20

RESULT: VARIANTS DETECTED

DETECTED VARIANTS

Gene	HGVS (coding)	Protein change	VAF	Depth	Tier	Classification
TP53	NM_000546.6:c.524G>A	p.Arg175His	0.60%	6,640×	I	Pathogenic

VARIANT ANNOTATION

TP53 p.Arg175His | Pathogenic (Tier I)

TP53 p.Arg175His is the most frequently observed p53 mutation in human cancer. It disrupts the structural zinc-binding domain of the DNA-binding domain, abolishing wild-type tumour suppressor function, while also conferring dominant-negative and gain-of-function oncogenic properties including promotion of invasion, metastasis, and MDM2 antagonism (COSMIC ID COSM10648). In OSCC, TP53 mutations are present in 50–70% of cases and are associated with aggressive tumour behaviour, locoregional recurrence, and reduced sensitivity to platinum-based chemotherapy. The VAF of 0.60% in a surveillance swab warrants clinical correlation to exclude early local recurrence. Imaging and clinical re-evaluation are recommended.

GENES WITH NO PATHOGENIC OR LIKELY PATHOGENIC VARIANTS DETECTED

AJUBA, CASP8, CDKN2A, CUL3, DPYD, EGFR, EP300, FAT1, FBXW7, HRAS, KEAP1, KMT2D, KRAS, NFE2L2, NOTCH1, NSD1, PIK3CA, PIK3R1, PTEN, TERT

Variants of uncertain significance (Tier III) are not reported.

METHOD SUMMARY

Cell-free and cellular DNA extracted from oral swab is subjected to hybridization capture using the IDT xGen Duplex UMI panel targeting 251 genomic regions (~77 kbp) across 21 cancer-relevant genes. Libraries incorporate 9-bp Unique Molecular Identifiers (UMIs) in a 9M1S+T read structure on both R1 and R2. Libraries are sequenced on Illumina NextSeq (2×150 bp). After alignment to GRCh38 (canonical chromosomes), reads are grouped by UMI using fgbio (adjacency strategy, ≤1 edit distance), and molecular consensus sequences are called requiring ≥2 reads per UMI family. Somatic variants are called on the consensus BAM with VarDict at minimum allele frequency 0.001. Variants ≥0.2% consensus VAF with ≥2 consensus supporting reads are reported. UMI deduplication reduces sequencing noise by >800-fold. Run-level negative controls are included on every sequencing run.

LIMITATIONS

This assay detects somatic single nucleotide variants and small indels within 251 target regions (~77 kbp) at VAF ≥0.2% with validated sensitivity of 98.4% at ≥20,000× raw input depth (~4,000× consensus after UMI deduplication). Structural variants, copy number alterations, gene fusions, large indels, intronic variants outside defined capture regions, and variants in regions with low hybridization capture efficiency are not reliably assessed. A negative result does not exclude malignancy. Tumour heterogeneity, low tumour cell content, and specimen quality may affect variant allele fraction. Variants below the 0.2% VAF reporting threshold may be present but are not reported. At 20,000× raw sequencing depth (~4,000× mean consensus), approximately 0.3% of targets may fall below 1,000× consensus depth; per-target coverage is provided in the laboratory technical report. Results should be interpreted in the context of clinical and pathological findings by a qualified clinician.

Reported by: Dr. Wisdom, MD PhD, FCAP **Report date: 2026-05-20**

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

HOTSPOT SCREENING SUMMARY

Patient: Dickens, Charles J. **Accession:** NGS-2026-052103

Hotspots listed below were explicitly interrogated in addition to all variants above 0.2 % VAF with ≥2 consensus reads across all 251 targets. Coverage criteria (≥100× consensus depth) were met at all targets.

EGFR		Cov ✓
L858R	c.2573T>G	Not detected
T790M	c.2369C>T	Not detected
Exon 19 del	c.2235_2249del	Not detected

KRAS		Cov ✓
G12D	c.35G>T	Not detected
G12V	c.35G>A	Not detected
G12C	c.34G>T	Not detected
G13D	c.38G>A	Not detected

PIK3CA		Cov ✓
E542K	c.1624G>A	Not detected
E545K	c.1633G>A	Not detected
H1047R	c.3140A>G	Not detected

TERT		Cov ✓
C228T	c.-124C>T	Not detected
C250T	c.-146C>T	Not detected

HRAS		Cov ✓
G12V	c.35G>T	Not detected
G12D	c.35G>A	Not detected
Q61L	c.182A>T	Not detected
Q61R	c.182A>G	Not detected

NOTCH1		Cov ✓
P2415L	c.7244C>T	Not detected
R1699W	c.5095C>T	Not detected

TP53		Cov ✓
R175H	c.524G>A	● **DETECTED**
R248Q	c.743G>A	Not detected
R248W	c.742C>T	Not detected
R273H	c.818G>A	Not detected
R282W	c.844C>T	Not detected

REMAINING PANEL GENES — SEQUENCE-LEVEL DETECTION, NO PREDEFINED HOTSPOT LIST

AJUBA · CASP8 · CDKN2A · CUL3 · DPYD · EP300 · FAT1 · FBXW7 · KEAP1 · KMT2D · NFE2L2 · NSD1 · PIK3R1 · PTEN

All variants above 0.2 % VAF with ≥2 consensus reads are reported for these genes. No variants meeting reporting criteria were detected. All targets in these genes met minimum coverage criteria (≥100× consensus depth).

Reported by: Dr. Wisdom, MD PhD, FCAP **Report date: 2026-05-20**

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

FULL PANEL COVERAGE REPORT — Dickens, Charles J. (NGS-2026-052103)

251 targets, 77,417 bp total. Mean consensus depth: 3,749×. Targets below 100× consensus: 0 / 251 (0%). Depths are derived from the consensus BAM (UMI deduplication, min-reads ≥ 2). Positions reported 1-based.

Gene	chr:start–end (1-based)	Len (bp)	Depth	Coverage status
DPYD				
DPYD	chr1:97,082,291–97,082,491	201	1,772×	✓ Covered
DPYD	chr1:97,449,958–97,450,158	201	3,609×	✓ Covered
DPYD	chr1:97,515,687–97,515,887	201	792×	✓ Covered
DPYD	chr1:97,573,763–97,573,963	201	5,897×	✓ Covered
PTEN				
PTEN	chr10:87,864,450–87,864,568	119	848×	✓ Covered
PTEN	chr10:87,894,005–87,894,129	125	526×	✓ Covered
PTEN	chr10:87,925,493–87,925,577	85	2,685×	✓ Covered
PTEN	chr10:87,931,026–87,931,109	84	2,594×	✓ Covered
PTEN	chr10:87,932,993–87,933,271	279	2,988×	✓ Covered
PTEN	chr10:87,952,098–87,952,279	182	11,743×	✓ Covered
PTEN	chr10:87,957,833–87,958,039	207	1,315×	✓ Covered
PTEN	chr10:87,960,874–87,961,138	265	531×	✓ Covered
PTEN	chr10:87,965,267–87,965,489	223	3,802×	✓ Covered
HRAS				
HRAS	chr11:532,619–532,775	157	1,983×	✓ Covered
HRAS	chr11:533,433–533,632	200	6,894×	✓ Covered
HRAS	chr11:533,746–533,964	219	7,504×	✓ Covered
HRAS	chr11:534,192–534,342	151	13,071×	✓ Covered
KRAS				
KRAS	chr12:25,209,778–25,209,931	154	724×	✓ Covered
KRAS	chr12:25,225,594–25,225,793	200	10,065×	✓ Covered
KRAS	chr12:25,227,214–25,227,432	219	1,904×	✓ Covered
KRAS	chr12:25,245,254–25,245,404	151	5,803×	✓ Covered
KMT2D				
KMT2D	chr12:49,021,763–49,021,892	130	2,654×	✓ Covered
KMT2D	chr12:49,022,023–49,022,171	149	769×	✓ Covered
KMT2D	chr12:49,022,260–49,022,373	114	15,690×	✓ Covered
KMT2D	chr12:49,022,570–49,022,895	326	1,186×	✓ Covered
KMT2D	chr12:49,024,558–49,024,728	171	528×	✓ Covered
KMT2D	chr12:49,024,790–49,024,966	177	611×	✓ Covered
KMT2D	chr12:49,026,162–49,027,342	1,181	869×	✓ Covered
KMT2D	chr12:49,027,783–49,027,950	168	2,777×	✓ Covered
KMT2D	chr12:49,027,989–49,028,161	173	5,551×	✓ Covered
KMT2D	chr12:49,028,808–49,028,978	171	1,298×	✓ Covered
KMT2D	chr12:49,029,041–49,029,256	216	7,800×	✓ Covered
KMT2D	chr12:49,029,381–49,029,496	116	2,499×	✓ Covered
KMT2D	chr12:49,030,260–49,030,459	200	975×	✓ Covered
KMT2D	chr12:49,030,581–49,030,788	208	5,691×	✓ Covered
KMT2D	chr12:49,030,873–49,031,053	181	19,585×	✓ Covered
KMT2D	chr12:49,031,155–49,033,984	2,830	1,039×	✓ Covered
KMT2D	chr12:49,034,047–49,034,319	273	846×	✓ Covered
KMT2D	chr12:49,034,390–49,034,496	107	3,267×	✓ Covered
KMT2D	chr12:49,034,562–49,034,686	125	1,519×	✓ Covered
KMT2D	chr12:49,034,792–49,034,955	164	8,397×	✓ Covered
KMT2D	chr12:49,037,105–49,039,009	1,905	6,980×	✓ Covered
KMT2D	chr12:49,039,202–49,039,378	177	8,894×	✓ Covered
KMT2D	chr12:49,039,415–49,039,637	223	1,334×	✓ Covered

FULL PANEL COVERAGE REPORT — Dickens, Charles J. (NGS-2026-052103)

Gene	chr:start–end (1-based)	Len (bp)	Depth	Coverage status
KMT2D	chr12:49,039,704–49,041,555	1,852	4,094×	✓ Covered
KMT2D	chr12:49,041,635–49,041,725	91	1,173×	✓ Covered
KMT2D	chr12:49,041,897–49,042,010	114	219×	✓ Covered
KMT2D	chr12:49,042,069–49,042,350	282	5,069×	✓ Covered
KMT2D	chr12:49,042,541–49,042,665	125	839×	✓ Covered
KMT2D	chr12:49,042,721–49,042,898	178	861×	✓ Covered
KMT2D	chr12:49,043,056–49,043,206	151	1,065×	✓ Covered
KMT2D	chr12:49,043,343–49,043,448	106	1,920×	✓ Covered
KMT2D	chr12:49,043,615–49,043,802	188	776×	✓ Covered
KMT2D	chr12:49,043,848–49,044,018	171	1,607×	✓ Covered
KMT2D	chr12:49,044,180–49,044,324	145	2,611×	✓ Covered
KMT2D	chr12:49,044,383–49,044,542	160	1,441×	✓ Covered
KMT2D	chr12:49,044,724–49,044,985	262	3,250×	✓ Covered
KMT2D	chr12:49,045,900–49,045,987	88	1,252×	✓ Covered
KMT2D	chr12:49,046,045–49,046,194	150	135×	✓ Covered
KMT2D	chr12:49,046,240–49,046,444	205	2,740×	✓ Covered
KMT2D	chr12:49,046,589–49,046,810	222	957×	✓ Covered
KMT2D	chr12:49,047,945–49,048,089	145	2,655×	✓ Covered
KMT2D	chr12:49,048,639–49,048,789	151	2,023×	✓ Covered
KMT2D	chr12:49,049,085–49,049,238	154	2,950×	✓ Covered
KMT2D	chr12:49,049,662–49,050,810	1,149	7,807×	✓ Covered
KMT2D	chr12:49,050,866–49,052,444	1,579	694×	✓ Covered
KMT2D	chr12:49,052,544–49,052,729	186	12,846×	✓ Covered
KMT2D	chr12:49,052,895–49,053,092	198	16,900×	✓ Covered
KMT2D	chr12:49,053,187–49,053,341	155	866×	✓ Covered
KMT2D	chr12:49,053,456–49,053,661	206	824×	✓ Covered
KMT2D	chr12:49,053,958–49,054,160	203	1,478×	✓ Covered
KMT2D	chr12:49,054,287–49,054,436	150	7,026×	✓ Covered
KMT2D	chr12:49,054,508–49,054,771	264	2,242×	✓ Covered
KMT2D	chr12:49,054,880–49,055,046	167	2,551×	✓ Covered
KMT2D	chr12:49,055,256–49,055,344	89	3,025×	✓ Covered
AJUBA				
AJUBA	chr14:22,973,426–22,973,588	163	33,482×	✓ Covered
AJUBA	chr14:22,974,027–22,974,135	109	2,134×	✓ Covered
AJUBA	chr14:22,974,819–22,974,910	92	3,284×	✓ Covered
AJUBA	chr14:22,974,954–22,975,124	171	882×	✓ Covered
AJUBA	chr14:22,976,436–22,976,538	103	1,841×	✓ Covered
AJUBA	chr14:22,976,625–22,976,732	108	5,416×	✓ Covered
AJUBA	chr14:22,978,324–22,978,465	142	235×	✓ Covered
AJUBA	chr14:22,981,241–22,982,286	1,046	3,466×	✓ Covered
TP53				
TP53	chr17:7,669,592–7,669,710	119	2,585×	✓ Covered
TP53	chr17:7,670,589–7,670,735	147	486×	✓ Covered
TP53	chr17:7,673,515–7,673,628	114	1,773×	✓ Covered
TP53	chr17:7,673,681–7,673,857	177	2,377×	✓ Covered
TP53	chr17:7,674,161–7,674,310	150	1,209×	✓ Covered
TP53	chr17:7,674,839–7,674,991	153	453×	✓ Covered
TP53	chr17:7,675,033–7,675,256	224	2,540×	● **VARIANT DETECTED**
TP53	chr17:7,675,974–7,676,292	319	1,421×	✓ Covered
TP53	chr17:7,676,362–7,676,423	62	8,054×	✓ Covered
TP53	chr17:7,676,501–7,676,614	114	599×	✓ Covered

FULL PANEL COVERAGE REPORT — Dickens, Charles J. (NGS-2026-052103)

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KEAP1				
KEAP1	chr19:10,486,635–10,486,838	204	1,714×	✓ Covered
KEAP1	chr19:10,489,172–10,489,388	217	1,932×	✓ Covered
KEAP1	chr19:10,489,628–10,489,873	246	1,054×	✓ Covered
KEAP1	chr19:10,491,557–10,492,282	726	6,609×	✓ Covered
KEAP1	chr19:10,499,375–10,500,053	679	4,480×	✓ Covered
NFE2L2				
NFE2L2	chr2:177,233,985–177,234,291	307	4,814×	✓ Covered
CASP8				
CASP8	chr2:201,266,467–201,266,811	345	5,047×	✓ Covered
CASP8	chr2:201,271,496–201,271,641	146	4,534×	✓ Covered
CASP8	chr2:201,272,618–201,272,796	179	471×	✓ Covered
CASP8	chr2:201,272,878–201,272,962	85	5,240×	✓ Covered
CASP8	chr2:201,274,869–201,274,973	105	7,252×	✓ Covered
CASP8	chr2:201,276,807–201,276,988	182	4,610×	✓ Covered
CASP8	chr2:201,284,796–201,285,337	542	11,616×	✓ Covered
CASP8	chr2:201,286,439–201,286,611	173	8,417×	✓ Covered
CUL3				
CUL3	chr2:224,474,228–224,474,396	169	1,557×	✓ Covered
CUL3	chr2:224,478,180–224,478,365	186	4,124×	✓ Covered
CUL3	chr2:224,481,872–224,482,098	227	917×	✓ Covered
CUL3	chr2:224,495,812–224,495,986	175	114×	✓ Covered
CUL3	chr2:224,497,733–224,497,869	137	936×	✓ Covered
CUL3	chr2:224,500,343–224,500,507	165	19,955×	✓ Covered
CUL3	chr2:224,502,945–224,503,092	148	348×	✓ Covered
CUL3	chr2:224,503,632–224,503,842	211	521×	✓ Covered
CUL3	chr2:224,505,936–224,506,152	217	2,715×	✓ Covered
CUL3	chr2:224,506,838–224,507,023	186	3,143×	✓ Covered
CUL3	chr2:224,511,334–224,511,602	269	1,004×	✓ Covered
CUL3	chr2:224,513,504–224,513,658	155	833×	✓ Covered
CUL3	chr2:224,514,592–224,514,792	201	441×	✓ Covered
CUL3	chr2:224,535,508–224,535,661	154	2,635×	✓ Covered
CUL3	chr2:224,557,639–224,557,876	238	5,386×	✓ Covered
CUL3	chr2:224,584,924–224,585,029	106	1,607×	✓ Covered
EP300				
EP300	chr22:41,092,985–41,093,118	134	8,282×	✓ Covered
EP300	chr22:41,117,167–41,117,841	675	274×	✓ Covered
EP300	chr22:41,125,844–41,126,060	217	890×	✓ Covered
EP300	chr22:41,127,467–41,127,768	302	879×	✓ Covered
EP300	chr22:41,129,870–41,130,023	154	931×	✓ Covered
EP300	chr22:41,131,368–41,131,653	286	1,332×	✓ Covered
EP300	chr22:41,135,793–41,135,926	134	837×	✓ Covered
EP300	chr22:41,137,633–41,137,810	178	10,929×	✓ Covered
EP300	chr22:41,140,120–41,140,277	158	1,136×	✓ Covered
EP300	chr22:41,141,028–41,141,242	215	805×	✓ Covered
EP300	chr22:41,146,719–41,146,836	118	10,764×	✓ Covered
EP300	chr22:41,147,817–41,147,966	150	375×	✓ Covered
EP300	chr22:41,149,018–41,149,195	178	197×	✓ Covered
EP300	chr22:41,149,741–41,150,218	478	528×	✓ Covered
EP300	chr22:41,151,813–41,152,032	220	521×	✓ Covered
EP300	chr22:41,152,186–41,152,370	185	1,742×	✓ Covered

FULL PANEL COVERAGE REPORT — Dickens, Charles J. (NGS-2026-052103)

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
EP300	chr22:41,154,975–41,155,133	159	2,913×	✓ Covered
EP300	chr22:41,157,149–41,157,428	280	892×	✓ Covered
EP300	chr22:41,158,392–41,158,520	129	6,927×	✓ Covered
EP300	chr22:41,160,622–41,160,742	121	13,461×	✓ Covered
EP300	chr22:41,162,703–41,162,799	97	1,031×	✓ Covered
EP300	chr22:41,164,033–41,164,150	118	250×	✓ Covered
EP300	chr22:41,166,579–41,166,686	108	3,794×	✓ Covered
EP300	chr22:41,168,429–41,168,619	191	2,610×	✓ Covered
EP300	chr22:41,168,701–41,168,887	187	361×	✓ Covered
EP300	chr22:41,169,483–41,169,636	154	2,317×	✓ Covered
EP300	chr22:41,170,386–41,170,591	206	730×	✓ Covered
EP300	chr22:41,172,479–41,172,683	205	3,641×	✓ Covered
EP300	chr22:41,173,603–41,173,804	202	13,435×	✓ Covered
EP300	chr22:41,176,227–41,176,548	322	6,685×	✓ Covered
EP300	chr22:41,176,753–41,178,973	2,221	20,101×	✓ Covered
PIK3CA				
PIK3CA	chr3:179,218,190–179,218,354	165	604×	✓ Covered
PIK3CA	chr3:179,234,074–179,234,381	308	6,094×	✓ Covered
FBXW7				
FBXW7	chr4:152,325,986–152,326,251	266	862×	✓ Covered
FAT1				
FAT1	chr4:186,588,575–186,589,240	666	671×	✓ Covered
FAT1	chr4:186,595,669–186,595,846	178	8,043×	✓ Covered
FAT1	chr4:186,596,520–186,597,191	672	738×	✓ Covered
FAT1	chr4:186,597,662–186,597,812	151	2,496×	✓ Covered
FAT1	chr4:186,597,952–186,598,145	194	1,128×	✓ Covered
FAT1	chr4:186,599,878–186,600,380	503	5,001×	✓ Covered
FAT1	chr4:186,601,249–186,601,446	198	1,093×	✓ Covered
FAT1	chr4:186,602,883–186,603,054	172	2,848×	✓ Covered
FAT1	chr4:186,603,156–186,603,997	842	2,464×	✓ Covered
FAT1	chr4:186,604,357–186,604,594	238	3,834×	✓ Covered
FAT1	chr4:186,606,050–186,606,233	184	2,921×	✓ Covered
FAT1	chr4:186,609,163–186,609,340	178	251×	✓ Covered
FAT1	chr4:186,609,781–186,610,035	255	1,519×	✓ Covered
FAT1	chr4:186,611,366–186,611,795	430	1,260×	✓ Covered
FAT1	chr4:186,613,089–186,613,362	274	249×	✓ Covered
FAT1	chr4:186,614,171–186,614,364	194	1,326×	✓ Covered
FAT1	chr4:186,616,985–186,617,221	237	2,950×	✓ Covered
FAT1	chr4:186,617,688–186,621,795	4,108	2,171×	✓ Covered
FAT1	chr4:186,628,134–186,628,384	251	1,287×	✓ Covered
FAT1	chr4:186,628,468–186,628,783	316	2,566×	✓ Covered
FAT1	chr4:186,633,664–186,633,843	180	1,092×	✓ Covered
FAT1	chr4:186,636,005–186,636,255	251	3,919×	✓ Covered
FAT1	chr4:186,636,565–186,636,934	370	754×	✓ Covered
FAT1	chr4:186,639,702–186,639,803	102	407×	✓ Covered
FAT1	chr4:186,663,279–186,663,633	355	2,899×	✓ Covered
FAT1	chr4:186,706,543–186,709,847	3,305	3,028×	✓ Covered
TERT				
TERT	chr5:1,295,050–1,295,250	201	52,211×	✓ Covered
PIK3R1				
PIK3R1	chr5:68,294,516–68,294,698	183	4,286×	✓ Covered

FULL PANEL COVERAGE REPORT — Dickens, Charles J. (NGS-2026-052103)

Gene	chr:start–end (1-based)	Len (bp)	Depth	Coverage status
PIK3R1	chr5:68,295,128–68,295,344	217	13,246×	✓ Covered
PIK3R1	chr5:68,295,400–68,295,508	109	1,972×	✓ Covered
NSD1				
NSD1	chr5:177,135,084–177,136,050	967	11,368×	✓ Covered
NSD1	chr5:177,191,864–177,192,039	176	892×	✓ Covered
NSD1	chr5:177,204,100–177,204,312	213	7,892×	✓ Covered
NSD1	chr5:177,209,616–177,212,215	2,600	4,211×	✓ Covered
NSD1	chr5:177,235,801–177,235,965	165	4,291×	✓ Covered
NSD1	chr5:177,238,217–177,238,527	311	215×	✓ Covered
NSD1	chr5:177,239,736–177,239,885	150	10,041×	✓ Covered
NSD1	chr5:177,244,175–177,244,290	116	1,820×	✓ Covered
NSD1	chr5:177,246,658–177,246,816	159	1,110×	✓ Covered
NSD1	chr5:177,248,161–177,248,344	184	6,096×	✓ Covered
NSD1	chr5:177,251,710–177,251,873	164	924×	✓ Covered
NSD1	chr5:177,256,931–177,257,171	241	3,868×	✓ Covered
NSD1	chr5:177,259,969–177,260,188	220	638×	✓ Covered
NSD1	chr5:177,267,542–177,267,738	197	3,464×	✓ Covered
NSD1	chr5:177,269,582–177,269,827	246	1,972×	✓ Covered
NSD1	chr5:177,273,652–177,273,804	153	2,003×	✓ Covered
NSD1	chr5:177,280,545–177,280,854	310	2,379×	✓ Covered
NSD1	chr5:177,282,445–177,282,601	157	827×	✓ Covered
NSD1	chr5:177,283,767–177,283,948	182	521×	✓ Covered
NSD1	chr5:177,288,799–177,288,945	147	11,881×	✓ Covered
NSD1	chr5:177,291,934–177,292,178	245	5,051×	✓ Covered
NSD1	chr5:177,293,812–177,295,476	1,665	1,672×	✓ Covered
EGFR				
EGFR	chr7:55,173,901–55,174,063	163	6,131×	✓ Covered
EGFR	chr7:55,174,702–55,174,840	139	591×	✓ Covered
EGFR	chr7:55,181,273–55,181,498	226	969×	✓ Covered
EGFR	chr7:55,191,699–55,191,894	196	591×	✓ Covered
CDKN2A				
CDKN2A	chr9:21,968,212–21,968,262	51	13,455×	✓ Covered
CDKN2A	chr9:21,970,882–21,971,228	347	1,391×	✓ Covered
CDKN2A	chr9:21,974,658–21,974,846	189	11,730×	✓ Covered
CDKN2A	chr9:21,974,658–21,974,847	190	1,175×	✓ Covered
NOTCH1				
NOTCH1	chr9:136,496,054–136,497,578	1,525	2,179×	✓ Covered
NOTCH1	chr9:136,498,879–136,499,016	138	588×	✓ Covered
NOTCH1	chr9:136,499,092–136,499,279	188	5,103×	✓ Covered
NOTCH1	chr9:136,500,532–136,500,867	336	3,556×	✓ Covered
NOTCH1	chr9:136,501,728–136,501,933	206	1,325×	✓ Covered
NOTCH1	chr9:136,501,981–136,502,108	128	2,309×	✓ Covered
NOTCH1	chr9:136,502,252–136,502,508	257	1,513×	✓ Covered
NOTCH1	chr9:136,503,162–136,503,350	189	6,223×	✓ Covered
NOTCH1	chr9:136,504,653–136,505,124	472	4,232×	✓ Covered
NOTCH1	chr9:136,505,290–136,505,901	612	312×	✓ Covered
NOTCH1	chr9:136,506,507–136,506,659	153	5,717×	✓ Covered
NOTCH1	chr9:136,506,696–136,506,993	298	6,155×	✓ Covered
NOTCH1	chr9:136,507,285–136,507,457	173	421×	✓ Covered
NOTCH1	chr9:136,507,935–136,508,159	225	3,632×	✓ Covered
NOTCH1	chr9:136,508,212–136,508,405	194	3,438×	✓ Covered

FULL PANEL COVERAGE REPORT — Dickens, Charles J. (NGS-2026-052103)

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
NOTCH1	chr9:136,508,850–136,509,091	242	1,937×	✓ Covered
NOTCH1	chr9:136,509,713–136,509,981	269	444×	✓ Covered
NOTCH1	chr9:136,510,633–136,510,825	193	822×	✓ Covered
NOTCH1	chr9:136,511,132–136,511,291	160	1,446×	✓ Covered
NOTCH1	chr9:136,513,001–136,513,154	154	2,374×	✓ Covered
NOTCH1	chr9:136,513,372–136,513,557	186	710×	✓ Covered
NOTCH1	chr9:136,514,490–136,514,722	233	272×	✓ Covered
NOTCH1	chr9:136,515,270–136,515,420	151	9,350×	✓ Covered
NOTCH1	chr9:136,515,463–136,515,736	274	1,930×	✓ Covered
NOTCH1	chr9:136,515,961–136,516,114	154	3,519×	✓ Covered
NOTCH1	chr9:136,517,252–136,517,405	154	3,759×	✓ Covered
NOTCH1	chr9:136,517,732–136,517,957	226	1,514×	✓ Covered
NOTCH1	chr9:136,518,117–136,518,312	196	2,440×	✓ Covered
NOTCH1	chr9:136,518,571–136,518,844	274	2,796×	✓ Covered
NOTCH1	chr9:136,519,423–136,519,585	163	1,416×	✓ Covered
NOTCH1	chr9:136,522,830–136,523,208	379	10,244×	✓ Covered
NOTCH1	chr9:136,523,697–136,523,999	303	338×	✓ Covered
NOTCH1	chr9:136,544,004–136,544,122	119	1,800×	✓ Covered
NOTCH1	chr9:136,545,706–136,545,806	101	4,352×	✓ Covered

PATIENT & SPECIMEN INFORMATION

Patient name	Kafka, Franz J.	**Accession**	NGS-2026-052105
Date of birth	1980-02-17 (Age 46)	**Specimen**	Oral swab, anterior tongue
MRN	50789065	**Collected**	2026-05-19
Ordering physician	Dr. Wisdom, MD	**Received**	2026-05-19
Clinical indication	Oral lesion — anterior tongue (surveillance post-excision)	**Reported**	2026-05-20

RESULT: NO REPORTABLE VARIANTS DETECTED

INTERPRETATION

No pathogenic or likely pathogenic somatic variants were detected in the 21 genes covered by this assay at ≥0.2% VAF with ≥2 consensus supporting reads. A negative result does not exclude malignancy. Variants below the reporting threshold, variants outside the 251 assay target regions, structural variants, and copy number alterations are not assessed by this assay. Clinical correlation is required; repeat testing or biopsy should be considered if clinical suspicion persists.

GENES WITH NO PATHOGENIC OR LIKELY PATHOGENIC VARIANTS DETECTED

AJUBA, CASP8, CDKN2A, CUL3, DPYD, EGFR, EP300, FAT1, FBXW7, HRAS, KEAP1, KMT2D, KRAS, NFE2L2, NOTCH1, NSD1, PIK3CA, PIK3R1, PTEN, TERT, TP53

Variants of uncertain significance (Tier III) are not reported.

METHOD SUMMARY

Cell-free and cellular DNA extracted from oral swab is subjected to hybridization capture using the IDT xGen Duplex UMI panel targeting 251 genomic regions (~77 kbp) across 21 cancer-relevant genes. Libraries incorporate 9-bp Unique Molecular Identifiers (UMIs) in a 9M1S+T read structure on both R1 and R2. Libraries are sequenced on Illumina NextSeq (2×150 bp). After alignment to GRCh38 (canonical chromosomes), reads are grouped by UMI using fgbio (adjacency strategy, ≤1 edit distance), and molecular consensus sequences are called requiring ≥2 reads per UMI family. Somatic variants are called on the consensus BAM with VarDict at minimum allele frequency 0.001. Variants ≥0.2% consensus VAF with ≥2 consensus supporting reads are reported. UMI deduplication reduces sequencing noise by >800-fold. Run-level negative controls are included on every sequencing run.

LIMITATIONS

This assay detects somatic single nucleotide variants and small indels within 251 target regions (~77 kbp) at VAF ≥0.2% with validated sensitivity of 98.4% at ≥20,000× raw input depth (~4,000× consensus after UMI deduplication). Structural variants, copy number alterations, gene fusions, large indels, intronic variants outside defined capture regions, and variants in regions with low hybridization capture efficiency are not reliably assessed. A negative result does not exclude malignancy. Tumour heterogeneity, low tumour cell content, and specimen quality may affect variant allele fraction. Variants below the 0.2% VAF reporting threshold may be present but are not reported. At 20,000× raw sequencing depth (~4,000× mean consensus), approximately 0.3% of targets may fall below 1,000× consensus depth; per-target coverage is provided in the laboratory technical report. Results should be interpreted in the context of clinical and pathological findings by a qualified clinician.

Reported by: Dr. Wisdom, MD PhD, FCAP **Report date: 2026-05-20**

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

HOTSPOT SCREENING SUMMARY

Patient: Kafka, Franz J. **Accession:** NGS-2026-052105

Hotspots listed below were explicitly interrogated in addition to all variants above 0.2 % VAF with ≥2 consensus reads across all 251 targets. Coverage criteria (≥100× consensus depth) were met at all targets.

EGFR		Cov ✓
L858R	c.2573T>G	Not detected
T790M	c.2369C>T	Not detected
Exon 19 del	c.2235_2249del	Not detected

KRAS		Cov ✓
G12D	c.35G>T	Not detected
G12V	c.35G>A	Not detected
G12C	c.34G>T	Not detected
G13D	c.38G>A	Not detected

PIK3CA		Cov ✓
E542K	c.1624G>A	Not detected
E545K	c.1633G>A	Not detected
H1047R	c.3140A>G	Not detected

TERT		Cov ✓
C228T	c.-124C>T	Not detected
C250T	c.-146C>T	Not detected

HRAS		Cov ✓
G12V	c.35G>T	Not detected
G12D	c.35G>A	Not detected
Q61L	c.182A>T	Not detected
Q61R	c.182A>G	Not detected

NOTCH1		Cov ✓
P2415L	c.7244C>T	Not detected
R1699W	c.5095C>T	Not detected

TP53		Cov ✓
R175H	c.524G>A	Not detected
R248Q	c.743G>A	Not detected
R248W	c.742C>T	Not detected
R273H	c.818G>A	Not detected
R282W	c.844C>T	Not detected

REMAINING PANEL GENES — SEQUENCE-LEVEL DETECTION, NO PREDEFINED HOTSPOT LIST

AJUBA · CASP8 · CDKN2A · CUL3 · DPYD · EP300 · FAT1 · FBXW7 · KEAP1 · KMT2D · NFE2L2 · NSD1 · PIK3R1 · PTEN

All variants above 0.2 % VAF with ≥2 consensus reads are reported for these genes. No variants meeting reporting criteria were detected. All targets in these genes met minimum coverage criteria (≥100× consensus depth).

Reported by: Dr. Wisdom, MD PhD, FCAP **Report date: 2026-05-20**

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

FULL PANEL COVERAGE REPORT — Kafka, Franz J. (NGS-2026-052105)

251 targets, 77,417 bp total. Mean consensus depth: 4,250×. Targets below 100× consensus: 1 / 251 (0%). Depths are derived from the consensus BAM (UMI deduplication, min-reads ≥ 2). Positions reported 1-based.

Gene	chr:start–end (1-based)	Len (bp)	Depth	Coverage status
DPYD				
DPYD	chr1:97,082,291–97,082,491	201	625×	✓ Covered
DPYD	chr1:97,449,958–97,450,158	201	1,663×	✓ Covered
DPYD	chr1:97,515,687–97,515,887	201	1,954×	✓ Covered
DPYD	chr1:97,573,763–97,573,963	201	12,503×	✓ Covered
PTEN				
PTEN	chr10:87,864,450–87,864,568	119	1,903×	✓ Covered
PTEN	chr10:87,894,005–87,894,129	125	8,192×	✓ Covered
PTEN	chr10:87,925,493–87,925,577	85	5,776×	✓ Covered
PTEN	chr10:87,931,026–87,931,109	84	5,431×	✓ Covered
PTEN	chr10:87,932,993–87,933,271	279	3,873×	✓ Covered
PTEN	chr10:87,952,098–87,952,279	182	423×	✓ Covered
PTEN	chr10:87,957,833–87,958,039	207	2,238×	✓ Covered
PTEN	chr10:87,960,874–87,961,138	265	2,651×	✓ Covered
PTEN	chr10:87,965,267–87,965,489	223	4,571×	✓ Covered
HRAS				
HRAS	chr11:532,619–532,775	157	2,392×	✓ Covered
HRAS	chr11:533,433–533,632	200	616×	✓ Covered
HRAS	chr11:533,746–533,964	219	818×	✓ Covered
HRAS	chr11:534,192–534,342	151	547×	✓ Covered
KRAS				
KRAS	chr12:25,209,778–25,209,931	154	1,498×	✓ Covered
KRAS	chr12:25,225,594–25,225,793	200	2,218×	✓ Covered
KRAS	chr12:25,227,214–25,227,432	219	16,206×	✓ Covered
KRAS	chr12:25,245,254–25,245,404	151	432×	✓ Covered
KMT2D				
KMT2D	chr12:49,021,763–49,021,892	130	943×	✓ Covered
KMT2D	chr12:49,022,023–49,022,171	149	1,148×	✓ Covered
KMT2D	chr12:49,022,260–49,022,373	114	6,418×	✓ Covered
KMT2D	chr12:49,022,570–49,022,895	326	16,956×	✓ Covered
KMT2D	chr12:49,024,558–49,024,728	171	2,990×	✓ Covered
KMT2D	chr12:49,024,790–49,024,966	177	1,984×	✓ Covered
KMT2D	chr12:49,026,162–49,027,342	1,181	5,460×	✓ Covered
KMT2D	chr12:49,027,783–49,027,950	168	611×	✓ Covered
KMT2D	chr12:49,027,989–49,028,161	173	721×	✓ Covered
KMT2D	chr12:49,028,808–49,028,978	171	1,579×	✓ Covered
KMT2D	chr12:49,029,041–49,029,256	216	15,407×	✓ Covered
KMT2D	chr12:49,029,381–49,029,496	116	682×	✓ Covered
KMT2D	chr12:49,030,260–49,030,459	200	962×	✓ Covered
KMT2D	chr12:49,030,581–49,030,788	208	245×	✓ Covered
KMT2D	chr12:49,030,873–49,031,053	181	2,765×	✓ Covered
KMT2D	chr12:49,031,155–49,033,984	2,830	1,507×	✓ Covered
KMT2D	chr12:49,034,047–49,034,319	273	1,124×	✓ Covered
KMT2D	chr12:49,034,390–49,034,496	107	913×	✓ Covered
KMT2D	chr12:49,034,562–49,034,686	125	3,275×	✓ Covered
KMT2D	chr12:49,034,792–49,034,955	164	20,574×	✓ Covered
KMT2D	chr12:49,037,105–49,039,009	1,905	1,119×	✓ Covered
KMT2D	chr12:49,039,202–49,039,378	177	4,286×	✓ Covered
KMT2D	chr12:49,039,415–49,039,637	223	460×	✓ Covered

FULL PANEL COVERAGE REPORT — Kafka, Franz J. (NGS-2026-052105)

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KMT2D	chr12:49,039,704–49,041,555	1,852	24,596×	✓ Covered
KMT2D	chr12:49,041,635–49,041,725	91	17,448×	✓ Covered
KMT2D	chr12:49,041,897–49,042,010	114	4,959×	✓ Covered
KMT2D	chr12:49,042,069–49,042,350	282	11,987×	✓ Covered
KMT2D	chr12:49,042,541–49,042,665	125	2,491×	✓ Covered
KMT2D	chr12:49,042,721–49,042,898	178	2,662×	✓ Covered
KMT2D	chr12:49,043,056–49,043,206	151	3,876×	✓ Covered
KMT2D	chr12:49,043,343–49,043,448	106	20,820×	✓ Covered
KMT2D	chr12:49,043,615–49,043,802	188	1,793×	✓ Covered
KMT2D	chr12:49,043,848–49,044,018	171	2,566×	✓ Covered
KMT2D	chr12:49,044,180–49,044,324	145	9,198×	✓ Covered
KMT2D	chr12:49,044,383–49,044,542	160	4,378×	✓ Covered
KMT2D	chr12:49,044,724–49,044,985	262	10,202×	✓ Covered
KMT2D	chr12:49,045,900–49,045,987	88	9,901×	✓ Covered
KMT2D	chr12:49,046,045–49,046,194	150	692×	✓ Covered
KMT2D	chr12:49,046,240–49,046,444	205	1,145×	✓ Covered
KMT2D	chr12:49,046,589–49,046,810	222	2,374×	✓ Covered
KMT2D	chr12:49,047,945–49,048,089	145	1,891×	✓ Covered
KMT2D	chr12:49,048,639–49,048,789	151	3,232×	✓ Covered
KMT2D	chr12:49,049,085–49,049,238	154	558×	✓ Covered
KMT2D	chr12:49,049,662–49,050,810	1,149	4,297×	✓ Covered
KMT2D	chr12:49,050,866–49,052,444	1,579	3,635×	✓ Covered
KMT2D	chr12:49,052,544–49,052,729	186	2,673×	✓ Covered
KMT2D	chr12:49,052,895–49,053,092	198	4,668×	✓ Covered
KMT2D	chr12:49,053,187–49,053,341	155	867×	✓ Covered
KMT2D	chr12:49,053,456–49,053,661	206	1,887×	✓ Covered
KMT2D	chr12:49,053,958–49,054,160	203	1,122×	✓ Covered
KMT2D	chr12:49,054,287–49,054,436	150	280×	✓ Covered
KMT2D	chr12:49,054,508–49,054,771	264	12,446×	✓ Covered
KMT2D	chr12:49,054,880–49,055,046	167	875×	✓ Covered
KMT2D	chr12:49,055,256–49,055,344	89	2,613×	✓ Covered
AJUBA				
AJUBA	chr14:22,973,426–22,973,588	163	2,113×	✓ Covered
AJUBA	chr14:22,974,027–22,974,135	109	12,696×	✓ Covered
AJUBA	chr14:22,974,819–22,974,910	92	1,076×	✓ Covered
AJUBA	chr14:22,974,954–22,975,124	171	7,171×	✓ Covered
AJUBA	chr14:22,976,436–22,976,538	103	3,489×	✓ Covered
AJUBA	chr14:22,976,625–22,976,732	108	1,022×	✓ Covered
AJUBA	chr14:22,978,324–22,978,465	142	205×	✓ Covered
AJUBA	chr14:22,981,241–22,982,286	1,046	6,397×	✓ Covered
TP53				
TP53	chr17:7,669,592–7,669,710	119	9,892×	✓ Covered
TP53	chr17:7,670,589–7,670,735	147	1,208×	✓ Covered
TP53	chr17:7,673,515–7,673,628	114	2,103×	✓ Covered
TP53	chr17:7,673,681–7,673,857	177	4,895×	✓ Covered
TP53	chr17:7,674,161–7,674,310	150	3,979×	✓ Covered
TP53	chr17:7,674,839–7,674,991	153	2,561×	✓ Covered
TP53	chr17:7,675,033–7,675,256	224	2,396×	✓ Covered
TP53	chr17:7,675,974–7,676,292	319	2,506×	✓ Covered
TP53	chr17:7,676,362–7,676,423	62	3,470×	✓ Covered
TP53	chr17:7,676,501–7,676,614	114	1,322×	✓ Covered

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Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KEAP1				
KEAP1	chr19:10,486,635–10,486,838	204	693×	✓ Covered
KEAP1	chr19:10,489,172–10,489,388	217	1,231×	✓ Covered
KEAP1	chr19:10,489,628–10,489,873	246	1,784×	✓ Covered
KEAP1	chr19:10,491,557–10,492,282	726	3,600×	✓ Covered
KEAP1	chr19:10,499,375–10,500,053	679	2,194×	✓ Covered
NFE2L2				
NFE2L2	chr2:177,233,985–177,234,291	307	2,235×	✓ Covered
CASP8				
CASP8	chr2:201,266,467–201,266,811	345	1,657×	✓ Covered
CASP8	chr2:201,271,496–201,271,641	146	1,136×	✓ Covered
CASP8	chr2:201,272,618–201,272,796	179	3,439×	✓ Covered
CASP8	chr2:201,272,878–201,272,962	85	405×	✓ Covered
CASP8	chr2:201,274,869–201,274,973	105	3,837×	✓ Covered
CASP8	chr2:201,276,807–201,276,988	182	6,020×	✓ Covered
CASP8	chr2:201,284,796–201,285,337	542	1,216×	✓ Covered
CASP8	chr2:201,286,439–201,286,611	173	2,726×	✓ Covered
CUL3				
CUL3	chr2:224,474,228–224,474,396	169	3,362×	✓ Covered
CUL3	chr2:224,478,180–224,478,365	186	17,869×	✓ Covered
CUL3	chr2:224,481,872–224,482,098	227	1,319×	✓ Covered
CUL3	chr2:224,495,812–224,495,986	175	1,630×	✓ Covered
CUL3	chr2:224,497,733–224,497,869	137	2,058×	✓ Covered
CUL3	chr2:224,500,343–224,500,507	165	3,754×	✓ Covered
CUL3	chr2:224,502,945–224,503,092	148	1,954×	✓ Covered
CUL3	chr2:224,503,632–224,503,842	211	34,651×	✓ Covered
CUL3	chr2:224,505,936–224,506,152	217	838×	✓ Covered
CUL3	chr2:224,506,838–224,507,023	186	5,295×	✓ Covered
CUL3	chr2:224,511,334–224,511,602	269	1,268×	✓ Covered
CUL3	chr2:224,513,504–224,513,658	155	1,023×	✓ Covered
CUL3	chr2:224,514,592–224,514,792	201	3,441×	✓ Covered
CUL3	chr2:224,535,508–224,535,661	154	1,536×	✓ Covered
CUL3	chr2:224,557,639–224,557,876	238	1,810×	✓ Covered
CUL3	chr2:224,584,924–224,585,029	106	782×	✓ Covered
EP300				
EP300	chr22:41,092,985–41,093,118	134	4,057×	✓ Covered
EP300	chr22:41,117,167–41,117,841	675	1,924×	✓ Covered
EP300	chr22:41,125,844–41,126,060	217	236×	✓ Covered
EP300	chr22:41,127,467–41,127,768	302	8,428×	✓ Covered
EP300	chr22:41,129,870–41,130,023	154	1,157×	✓ Covered
EP300	chr22:41,131,368–41,131,653	286	7,585×	✓ Covered
EP300	chr22:41,135,793–41,135,926	134	1,779×	✓ Covered
EP300	chr22:41,137,633–41,137,810	178	363×	✓ Covered
EP300	chr22:41,140,120–41,140,277	158	1,734×	✓ Covered
EP300	chr22:41,141,028–41,141,242	215	417×	✓ Covered
EP300	chr22:41,146,719–41,146,836	118	16,257×	✓ Covered
EP300	chr22:41,147,817–41,147,966	150	792×	✓ Covered
EP300	chr22:41,149,018–41,149,195	178	13,908×	✓ Covered
EP300	chr22:41,149,741–41,150,218	478	1,007×	✓ Covered
EP300	chr22:41,151,813–41,152,032	220	1,178×	✓ Covered
EP300	chr22:41,152,186–41,152,370	185	3,296×	✓ Covered

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Gene	chr:start–end (1-based)	Len (bp)	Depth	Coverage status
EP300	chr22:41,154,975–41,155,133	159	2,884×	✓ Covered
EP300	chr22:41,157,149–41,157,428	280	3,445×	✓ Covered
EP300	chr22:41,158,392–41,158,520	129	2,370×	✓ Covered
EP300	chr22:41,160,622–41,160,742	121	3,797×	✓ Covered
EP300	chr22:41,162,703–41,162,799	97	15,367×	✓ Covered
EP300	chr22:41,164,033–41,164,150	118	9,210×	✓ Covered
EP300	chr22:41,166,579–41,166,686	108	**66×**	⚠ Below 100× threshold
EP300	chr22:41,168,429–41,168,619	191	9,105×	✓ Covered
EP300	chr22:41,168,701–41,168,887	187	1,188×	✓ Covered
EP300	chr22:41,169,483–41,169,636	154	10,683×	✓ Covered
EP300	chr22:41,170,386–41,170,591	206	1,572×	✓ Covered
EP300	chr22:41,172,479–41,172,683	205	1,133×	✓ Covered
EP300	chr22:41,173,603–41,173,804	202	1,076×	✓ Covered
EP300	chr22:41,176,227–41,176,548	322	3,473×	✓ Covered
EP300	chr22:41,176,753–41,178,973	2,221	430×	✓ Covered
PIK3CA				
PIK3CA	chr3:179,218,190–179,218,354	165	749×	✓ Covered
PIK3CA	chr3:179,234,074–179,234,381	308	591×	✓ Covered
FBXW7				
FBXW7	chr4:152,325,986–152,326,251	266	1,327×	✓ Covered
FAT1				
FAT1	chr4:186,588,575–186,589,240	666	605×	✓ Covered
FAT1	chr4:186,595,669–186,595,846	178	2,259×	✓ Covered
FAT1	chr4:186,596,520–186,597,191	672	925×	✓ Covered
FAT1	chr4:186,597,662–186,597,812	151	19,718×	✓ Covered
FAT1	chr4:186,597,952–186,598,145	194	282×	✓ Covered
FAT1	chr4:186,599,878–186,600,380	503	9,590×	✓ Covered
FAT1	chr4:186,601,249–186,601,446	198	868×	✓ Covered
FAT1	chr4:186,602,883–186,603,054	172	543×	✓ Covered
FAT1	chr4:186,603,156–186,603,997	842	4,076×	✓ Covered
FAT1	chr4:186,604,357–186,604,594	238	720×	✓ Covered
FAT1	chr4:186,606,050–186,606,233	184	9,420×	✓ Covered
FAT1	chr4:186,609,163–186,609,340	178	3,983×	✓ Covered
FAT1	chr4:186,609,781–186,610,035	255	850×	✓ Covered
FAT1	chr4:186,611,366–186,611,795	430	1,024×	✓ Covered
FAT1	chr4:186,613,089–186,613,362	274	1,115×	✓ Covered
FAT1	chr4:186,614,171–186,614,364	194	458×	✓ Covered
FAT1	chr4:186,616,985–186,617,221	237	868×	✓ Covered
FAT1	chr4:186,617,688–186,621,795	4,108	614×	✓ Covered
FAT1	chr4:186,628,134–186,628,384	251	7,328×	✓ Covered
FAT1	chr4:186,628,468–186,628,783	316	3,658×	✓ Covered
FAT1	chr4:186,633,664–186,633,843	180	5,352×	✓ Covered
FAT1	chr4:186,636,005–186,636,255	251	21,419×	✓ Covered
FAT1	chr4:186,636,565–186,636,934	370	1,214×	✓ Covered
FAT1	chr4:186,639,702–186,639,803	102	427×	✓ Covered
FAT1	chr4:186,663,279–186,663,633	355	20,293×	✓ Covered
FAT1	chr4:186,706,543–186,709,847	3,305	1,349×	✓ Covered
TERT				
TERT	chr5:1,295,050–1,295,250	201	2,665×	✓ Covered
PIK3R1				
PIK3R1	chr5:68,294,516–68,294,698	183	1,417×	✓ Covered

FULL PANEL COVERAGE REPORT — Kafka, Franz J. (NGS-2026-052105)

Gene	chr:start–end (1-based)	Len (bp)	Depth	Coverage status
PIK3R1	chr5:68,295,128–68,295,344	217	939×	✓ Covered
PIK3R1	chr5:68,295,400–68,295,508	109	2,801×	✓ Covered
NSD1				
NSD1	chr5:177,135,084–177,136,050	967	1,135×	✓ Covered
NSD1	chr5:177,191,864–177,192,039	176	8,835×	✓ Covered
NSD1	chr5:177,204,100–177,204,312	213	8,511×	✓ Covered
NSD1	chr5:177,209,616–177,212,215	2,600	305×	✓ Covered
NSD1	chr5:177,235,801–177,235,965	165	775×	✓ Covered
NSD1	chr5:177,238,217–177,238,527	311	4,751×	✓ Covered
NSD1	chr5:177,239,736–177,239,885	150	1,690×	✓ Covered
NSD1	chr5:177,244,175–177,244,290	116	165×	✓ Covered
NSD1	chr5:177,246,658–177,246,816	159	937×	✓ Covered
NSD1	chr5:177,248,161–177,248,344	184	9,697×	✓ Covered
NSD1	chr5:177,251,710–177,251,873	164	1,338×	✓ Covered
NSD1	chr5:177,256,931–177,257,171	241	2,875×	✓ Covered
NSD1	chr5:177,259,969–177,260,188	220	9,531×	✓ Covered
NSD1	chr5:177,267,542–177,267,738	197	564×	✓ Covered
NSD1	chr5:177,269,582–177,269,827	246	369×	✓ Covered
NSD1	chr5:177,273,652–177,273,804	153	225×	✓ Covered
NSD1	chr5:177,280,545–177,280,854	310	1,542×	✓ Covered
NSD1	chr5:177,282,445–177,282,601	157	2,358×	✓ Covered
NSD1	chr5:177,283,767–177,283,948	182	1,341×	✓ Covered
NSD1	chr5:177,288,799–177,288,945	147	2,388×	✓ Covered
NSD1	chr5:177,291,934–177,292,178	245	8,925×	✓ Covered
NSD1	chr5:177,293,812–177,295,476	1,665	35,320×	✓ Covered
EGFR				
EGFR	chr7:55,173,901–55,174,063	163	402×	✓ Covered
EGFR	chr7:55,174,702–55,174,840	139	972×	✓ Covered
EGFR	chr7:55,181,273–55,181,498	226	4,934×	✓ Covered
EGFR	chr7:55,191,699–55,191,894	196	656×	✓ Covered
CDKN2A				
CDKN2A	chr9:21,968,212–21,968,262	51	1,241×	✓ Covered
CDKN2A	chr9:21,970,882–21,971,228	347	4,075×	✓ Covered
CDKN2A	chr9:21,974,658–21,974,846	189	2,444×	✓ Covered
CDKN2A	chr9:21,974,658–21,974,847	190	4,080×	✓ Covered
NOTCH1				
NOTCH1	chr9:136,496,054–136,497,578	1,525	3,629×	✓ Covered
NOTCH1	chr9:136,498,879–136,499,016	138	205×	✓ Covered
NOTCH1	chr9:136,499,092–136,499,279	188	5,600×	✓ Covered
NOTCH1	chr9:136,500,532–136,500,867	336	5,110×	✓ Covered
NOTCH1	chr9:136,501,728–136,501,933	206	18,781×	✓ Covered
NOTCH1	chr9:136,501,981–136,502,108	128	4,532×	✓ Covered
NOTCH1	chr9:136,502,252–136,502,508	257	2,930×	✓ Covered
NOTCH1	chr9:136,503,162–136,503,350	189	1,147×	✓ Covered
NOTCH1	chr9:136,504,653–136,505,124	472	1,292×	✓ Covered
NOTCH1	chr9:136,505,290–136,505,901	612	8,785×	✓ Covered
NOTCH1	chr9:136,506,507–136,506,659	153	749×	✓ Covered
NOTCH1	chr9:136,506,696–136,506,993	298	1,649×	✓ Covered
NOTCH1	chr9:136,507,285–136,507,457	173	1,102×	✓ Covered
NOTCH1	chr9:136,507,935–136,508,159	225	4,235×	✓ Covered
NOTCH1	chr9:136,508,212–136,508,405	194	1,108×	✓ Covered

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Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
NOTCH1	chr9:136,508,850–136,509,091	242	21,728×	✓ Covered
NOTCH1	chr9:136,509,713–136,509,981	269	5,893×	✓ Covered
NOTCH1	chr9:136,510,633–136,510,825	193	2,836×	✓ Covered
NOTCH1	chr9:136,511,132–136,511,291	160	769×	✓ Covered
NOTCH1	chr9:136,513,001–136,513,154	154	921×	✓ Covered
NOTCH1	chr9:136,513,372–136,513,557	186	3,247×	✓ Covered
NOTCH1	chr9:136,514,490–136,514,722	233	1,349×	✓ Covered
NOTCH1	chr9:136,515,270–136,515,420	151	979×	✓ Covered
NOTCH1	chr9:136,515,463–136,515,736	274	3,425×	✓ Covered
NOTCH1	chr9:136,515,961–136,516,114	154	16,226×	✓ Covered
NOTCH1	chr9:136,517,252–136,517,405	154	9,235×	✓ Covered
NOTCH1	chr9:136,517,732–136,517,957	226	1,220×	✓ Covered
NOTCH1	chr9:136,518,117–136,518,312	196	5,880×	✓ Covered
NOTCH1	chr9:136,518,571–136,518,844	274	4,325×	✓ Covered
NOTCH1	chr9:136,519,423–136,519,585	163	840×	✓ Covered
NOTCH1	chr9:136,522,830–136,523,208	379	1,895×	✓ Covered
NOTCH1	chr9:136,523,697–136,523,999	303	1,793×	✓ Covered
NOTCH1	chr9:136,544,004–136,544,122	119	16,136×	✓ Covered
NOTCH1	chr9:136,545,706–136,545,806	101	1,573×	✓ Covered

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PATIENT & SPECIMEN INFORMATION

Patient name	Morrison, Toni E.	**Accession**	NGS-2026-052106
Date of birth	1955-10-09 (Age 70)	**Specimen**	Oral swab, left retromolar trigone
MRN	60890176	**Collected**	2026-05-19
Ordering physician	Dr. Wisdom, MD	**Received**	2026-05-19
Clinical indication	Oral lesion — left retromolar trigone	**Reported**	2026-05-20

RESULT: NO REPORTABLE VARIANTS DETECTED

INTERPRETATION

No pathogenic or likely pathogenic somatic variants were detected in the 21 genes covered by this assay at ≥0.2% VAF with ≥2 consensus supporting reads. A negative result does not exclude malignancy. Variants below the reporting threshold, variants outside the 251 assay target regions, structural variants, and copy number alterations are not assessed by this assay. Clinical correlation is required; repeat testing or biopsy should be considered if clinical suspicion persists.

GENES WITH NO PATHOGENIC OR LIKELY PATHOGENIC VARIANTS DETECTED

AJUBA, CASP8, CDKN2A, CUL3, DPYD, EGFR, EP300, FAT1, FBXW7, HRAS, KEAP1, KMT2D, KRAS, NFE2L2, NOTCH1, NSD1, PIK3CA, PIK3R1, PTEN, TERT, TP53

Variants of uncertain significance (Tier III) are not reported.

METHOD SUMMARY

Cell-free and cellular DNA extracted from oral swab is subjected to hybridization capture using the IDT xGen Duplex UMI panel targeting 251 genomic regions (~77 kbp) across 21 cancer-relevant genes. Libraries incorporate 9-bp Unique Molecular Identifiers (UMIs) in a 9M1S+T read structure on both R1 and R2. Libraries are sequenced on Illumina NextSeq (2×150 bp). After alignment to GRCh38 (canonical chromosomes), reads are grouped by UMI using fgbio (adjacency strategy, ≤1 edit distance), and molecular consensus sequences are called requiring ≥2 reads per UMI family. Somatic variants are called on the consensus BAM with VarDict at minimum allele frequency 0.001. Variants ≥0.2% consensus VAF with ≥2 consensus supporting reads are reported. UMI deduplication reduces sequencing noise by >800-fold. Run-level negative controls are included on every sequencing run.

LIMITATIONS

This assay detects somatic single nucleotide variants and small indels within 251 target regions (~77 kbp) at VAF ≥0.2% with validated sensitivity of 98.4% at ≥20,000× raw input depth (~4,000× consensus after UMI deduplication). Structural variants, copy number alterations, gene fusions, large indels, intronic variants outside defined capture regions, and variants in regions with low hybridization capture efficiency are not reliably assessed. A negative result does not exclude malignancy. Tumour heterogeneity, low tumour cell content, and specimen quality may affect variant allele fraction. Variants below the 0.2% VAF reporting threshold may be present but are not reported. At 20,000× raw sequencing depth (~4,000× mean consensus), approximately 0.3% of targets may fall below 1,000× consensus depth; per-target coverage is provided in the laboratory technical report. Results should be interpreted in the context of clinical and pathological findings by a qualified clinician.

Reported by: Dr. Wisdom, MD PhD, FCAP **Report date: 2026-05-20**

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

HOTSPOT SCREENING SUMMARY

Patient: Morrison, Toni E. **Accession:** NGS-2026-052106

Hotspots listed below were explicitly interrogated in addition to all variants above 0.2 % VAF with ≥2 consensus reads across all 251 targets. Coverage criteria (≥100× consensus depth) were met at all targets.

EGFR		Cov ✓
L858R	c.2573T>G	Not detected
T790M	c.2369C>T	Not detected
Exon 19 del	c.2235_2249del	Not detected

KRAS		Cov ✓
G12D	c.35G>T	Not detected
G12V	c.35G>A	Not detected
G12C	c.34G>T	Not detected
G13D	c.38G>A	Not detected

PIK3CA		Cov ✓
E542K	c.1624G>A	Not detected
E545K	c.1633G>A	Not detected
H1047R	c.3140A>G	Not detected

TERT		Cov ✓
C228T	c.-124C>T	Not detected
C250T	c.-146C>T	Not detected

HRAS		Cov ✓
G12V	c.35G>T	Not detected
G12D	c.35G>A	Not detected
Q61L	c.182A>T	Not detected
Q61R	c.182A>G	Not detected

NOTCH1		Cov ✓
P2415L	c.7244C>T	Not detected
R1699W	c.5095C>T	Not detected

TP53		Cov ✓
R175H	c.524G>A	Not detected
R248Q	c.743G>A	Not detected
R248W	c.742C>T	Not detected
R273H	c.818G>A	Not detected
R282W	c.844C>T	Not detected

REMAINING PANEL GENES — SEQUENCE-LEVEL DETECTION, NO PREDEFINED HOTSPOT LIST

AJUBA · CASP8 · CDKN2A · CUL3 · DPYD · EP300 · FAT1 · FBXW7 · KEAP1 · KMT2D · NFE2L2 · NSD1 · PIK3R1 · PTEN

All variants above 0.2 % VAF with ≥2 consensus reads are reported for these genes. No variants meeting reporting criteria were detected. All targets in these genes met minimum coverage criteria (≥100× consensus depth).

Reported by: Dr. Wisdom, MD PhD, FCAP **Report date: 2026-05-20**

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

FULL PANEL COVERAGE REPORT — Morrison, Toni E. (NGS-2026-052106)

251 targets, 77,417 bp total. Mean consensus depth: 3,528×. Targets below 100× consensus: 3 / 251 (1%). Depths are derived from the consensus BAM (UMI deduplication, min-reads ≥ 2). Positions reported 1-based.

Gene	chr:start–end (1-based)	Len (bp)	Depth	Coverage status
DPYD				
DPYD	chr1:97,082,291–97,082,491	201	4,798×	✓ Covered
DPYD	chr1:97,449,958–97,450,158	201	1,681×	✓ Covered
DPYD	chr1:97,515,687–97,515,887	201	671×	✓ Covered
DPYD	chr1:97,573,763–97,573,963	201	597×	✓ Covered
PTEN				
PTEN	chr10:87,864,450–87,864,568	119	333×	✓ Covered
PTEN	chr10:87,894,005–87,894,129	125	1,805×	✓ Covered
PTEN	chr10:87,925,493–87,925,577	85	717×	✓ Covered
PTEN	chr10:87,931,026–87,931,109	84	6,468×	✓ Covered
PTEN	chr10:87,932,993–87,933,271	279	536×	✓ Covered
PTEN	chr10:87,952,098–87,952,279	182	2,628×	✓ Covered
PTEN	chr10:87,957,833–87,958,039	207	3,165×	✓ Covered
PTEN	chr10:87,960,874–87,961,138	265	2,562×	✓ Covered
PTEN	chr10:87,965,267–87,965,489	223	2,807×	✓ Covered
HRAS				
HRAS	chr11:532,619–532,775	157	13,894×	✓ Covered
HRAS	chr11:533,433–533,632	200	913×	✓ Covered
HRAS	chr11:533,746–533,964	219	3,252×	✓ Covered
HRAS	chr11:534,192–534,342	151	3,751×	✓ Covered
KRAS				
KRAS	chr12:25,209,778–25,209,931	154	547×	✓ Covered
KRAS	chr12:25,225,594–25,225,793	200	3,579×	✓ Covered
KRAS	chr12:25,227,214–25,227,432	219	3,909×	✓ Covered
KRAS	chr12:25,245,254–25,245,404	151	4,576×	✓ Covered
KMT2D				
KMT2D	chr12:49,021,763–49,021,892	130	1,857×	✓ Covered
KMT2D	chr12:49,022,023–49,022,171	149	4,016×	✓ Covered
KMT2D	chr12:49,022,260–49,022,373	114	3,202×	✓ Covered
KMT2D	chr12:49,022,570–49,022,895	326	6,653×	✓ Covered
KMT2D	chr12:49,024,558–49,024,728	171	830×	✓ Covered
KMT2D	chr12:49,024,790–49,024,966	177	7,403×	✓ Covered
KMT2D	chr12:49,026,162–49,027,342	1,181	536×	✓ Covered
KMT2D	chr12:49,027,783–49,027,950	168	1,711×	✓ Covered
KMT2D	chr12:49,027,989–49,028,161	173	2,959×	✓ Covered
KMT2D	chr12:49,028,808–49,028,978	171	7,877×	✓ Covered
KMT2D	chr12:49,029,041–49,029,256	216	1,308×	✓ Covered
KMT2D	chr12:49,029,381–49,029,496	116	3,980×	✓ Covered
KMT2D	chr12:49,030,260–49,030,459	200	1,915×	✓ Covered
KMT2D	chr12:49,030,581–49,030,788	208	26,397×	✓ Covered
KMT2D	chr12:49,030,873–49,031,053	181	2,512×	✓ Covered
KMT2D	chr12:49,031,155–49,033,984	2,830	2,500×	✓ Covered
KMT2D	chr12:49,034,047–49,034,319	273	746×	✓ Covered
KMT2D	chr12:49,034,390–49,034,496	107	305×	✓ Covered
KMT2D	chr12:49,034,562–49,034,686	125	1,147×	✓ Covered
KMT2D	chr12:49,034,792–49,034,955	164	747×	✓ Covered
KMT2D	chr12:49,037,105–49,039,009	1,905	2,738×	✓ Covered
KMT2D	chr12:49,039,202–49,039,378	177	2,195×	✓ Covered
KMT2D	chr12:49,039,415–49,039,637	223	7,796×	✓ Covered

FULL PANEL COVERAGE REPORT — Morrison, Toni E. (NGS-2026-052106)

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KMT2D	chr12:49,039,704–49,041,555	1,852	2,944×	✓ Covered
KMT2D	chr12:49,041,635–49,041,725	91	1,115×	✓ Covered
KMT2D	chr12:49,041,897–49,042,010	114	5,762×	✓ Covered
KMT2D	chr12:49,042,069–49,042,350	282	1,397×	✓ Covered
KMT2D	chr12:49,042,541–49,042,665	125	1,328×	✓ Covered
KMT2D	chr12:49,042,721–49,042,898	178	1,349×	✓ Covered
KMT2D	chr12:49,043,056–49,043,206	151	6,653×	✓ Covered
KMT2D	chr12:49,043,343–49,043,448	106	284×	✓ Covered
KMT2D	chr12:49,043,615–49,043,802	188	1,465×	✓ Covered
KMT2D	chr12:49,043,848–49,044,018	171	1,920×	✓ Covered
KMT2D	chr12:49,044,180–49,044,324	145	2,078×	✓ Covered
KMT2D	chr12:49,044,383–49,044,542	160	3,413×	✓ Covered
KMT2D	chr12:49,044,724–49,044,985	262	722×	✓ Covered
KMT2D	chr12:49,045,900–49,045,987	88	1,760×	✓ Covered
KMT2D	chr12:49,046,045–49,046,194	150	2,827×	✓ Covered
KMT2D	chr12:49,046,240–49,046,444	205	3,248×	✓ Covered
KMT2D	chr12:49,046,589–49,046,810	222	17,571×	✓ Covered
KMT2D	chr12:49,047,945–49,048,089	145	2,149×	✓ Covered
KMT2D	chr12:49,048,639–49,048,789	151	3,583×	✓ Covered
KMT2D	chr12:49,049,085–49,049,238	154	2,570×	✓ Covered
KMT2D	chr12:49,049,662–49,050,810	1,149	250×	✓ Covered
KMT2D	chr12:49,050,866–49,052,444	1,579	2,618×	✓ Covered
KMT2D	chr12:49,052,544–49,052,729	186	1,699×	✓ Covered
KMT2D	chr12:49,052,895–49,053,092	198	2,174×	✓ Covered
KMT2D	chr12:49,053,187–49,053,341	155	9,712×	✓ Covered
KMT2D	chr12:49,053,456–49,053,661	206	1,707×	✓ Covered
KMT2D	chr12:49,053,958–49,054,160	203	2,038×	✓ Covered
KMT2D	chr12:49,054,287–49,054,436	150	106×	✓ Covered
KMT2D	chr12:49,054,508–49,054,771	264	2,187×	✓ Covered
KMT2D	chr12:49,054,880–49,055,046	167	50,573×	✓ Covered
KMT2D	chr12:49,055,256–49,055,344	89	1,925×	✓ Covered
AJUBA				
AJUBA	chr14:22,973,426–22,973,588	163	22,158×	✓ Covered
AJUBA	chr14:22,974,027–22,974,135	109	1,319×	✓ Covered
AJUBA	chr14:22,974,819–22,974,910	92	1,413×	✓ Covered
AJUBA	chr14:22,974,954–22,975,124	171	4,679×	✓ Covered
AJUBA	chr14:22,976,436–22,976,538	103	2,577×	✓ Covered
AJUBA	chr14:22,976,625–22,976,732	108	1,582×	✓ Covered
AJUBA	chr14:22,978,324–22,978,465	142	856×	✓ Covered
AJUBA	chr14:22,981,241–22,982,286	1,046	825×	✓ Covered
TP53				
TP53	chr17:7,669,592–7,669,710	119	13,338×	✓ Covered
TP53	chr17:7,670,589–7,670,735	147	964×	✓ Covered
TP53	chr17:7,673,515–7,673,628	114	1,531×	✓ Covered
TP53	chr17:7,673,681–7,673,857	177	1,258×	✓ Covered
TP53	chr17:7,674,161–7,674,310	150	1,065×	✓ Covered
TP53	chr17:7,674,839–7,674,991	153	3,088×	✓ Covered
TP53	chr17:7,675,033–7,675,256	224	9,823×	✓ Covered
TP53	chr17:7,675,974–7,676,292	319	971×	✓ Covered
TP53	chr17:7,676,362–7,676,423	62	1,090×	✓ Covered
TP53	chr17:7,676,501–7,676,614	114	1,262×	✓ Covered

FULL PANEL COVERAGE REPORT — Morrison, Toni E. (NGS-2026-052106)

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KEAP1				
KEAP1	chr19:10,486,635–10,486,838	204	2,096×	✓ Covered
KEAP1	chr19:10,489,172–10,489,388	217	1,983×	✓ Covered
KEAP1	chr19:10,489,628–10,489,873	246	3,283×	✓ Covered
KEAP1	chr19:10,491,557–10,492,282	726	1,057×	✓ Covered
KEAP1	chr19:10,499,375–10,500,053	679	9,941×	✓ Covered
NFE2L2				
NFE2L2	chr2:177,233,985–177,234,291	307	1,235×	✓ Covered
CASP8				
CASP8	chr2:201,266,467–201,266,811	345	**39×**	⚠ Below 100× threshold
CASP8	chr2:201,271,496–201,271,641	146	1,404×	✓ Covered
CASP8	chr2:201,272,618–201,272,796	179	880×	✓ Covered
CASP8	chr2:201,272,878–201,272,962	85	555×	✓ Covered
CASP8	chr2:201,274,869–201,274,973	105	8,337×	✓ Covered
CASP8	chr2:201,276,807–201,276,988	182	2,935×	✓ Covered
CASP8	chr2:201,284,796–201,285,337	542	745×	✓ Covered
CASP8	chr2:201,286,439–201,286,611	173	745×	✓ Covered
CUL3				
CUL3	chr2:224,474,228–224,474,396	169	566×	✓ Covered
CUL3	chr2:224,478,180–224,478,365	186	2,561×	✓ Covered
CUL3	chr2:224,481,872–224,482,098	227	505×	✓ Covered
CUL3	chr2:224,495,812–224,495,986	175	1,094×	✓ Covered
CUL3	chr2:224,497,733–224,497,869	137	1,798×	✓ Covered
CUL3	chr2:224,500,343–224,500,507	165	6,218×	✓ Covered
CUL3	chr2:224,502,945–224,503,092	148	2,706×	✓ Covered
CUL3	chr2:224,503,632–224,503,842	211	6,748×	✓ Covered
CUL3	chr2:224,505,936–224,506,152	217	1,489×	✓ Covered
CUL3	chr2:224,506,838–224,507,023	186	2,508×	✓ Covered
CUL3	chr2:224,511,334–224,511,602	269	8,802×	✓ Covered
CUL3	chr2:224,513,504–224,513,658	155	354×	✓ Covered
CUL3	chr2:224,514,592–224,514,792	201	3,154×	✓ Covered
CUL3	chr2:224,535,508–224,535,661	154	1,540×	✓ Covered
CUL3	chr2:224,557,639–224,557,876	238	3,906×	✓ Covered
CUL3	chr2:224,584,924–224,585,029	106	1,723×	✓ Covered
EP300				
EP300	chr22:41,092,985–41,093,118	134	7,483×	✓ Covered
EP300	chr22:41,117,167–41,117,841	675	2,969×	✓ Covered
EP300	chr22:41,125,844–41,126,060	217	15,259×	✓ Covered
EP300	chr22:41,127,467–41,127,768	302	701×	✓ Covered
EP300	chr22:41,129,870–41,130,023	154	6,970×	✓ Covered
EP300	chr22:41,131,368–41,131,653	286	12,240×	✓ Covered
EP300	chr22:41,135,793–41,135,926	134	2,512×	✓ Covered
EP300	chr22:41,137,633–41,137,810	178	2,681×	✓ Covered
EP300	chr22:41,140,120–41,140,277	158	3,027×	✓ Covered
EP300	chr22:41,141,028–41,141,242	215	9,451×	✓ Covered
EP300	chr22:41,146,719–41,146,836	118	548×	✓ Covered
EP300	chr22:41,147,817–41,147,966	150	1,757×	✓ Covered
EP300	chr22:41,149,018–41,149,195	178	1,337×	✓ Covered
EP300	chr22:41,149,741–41,150,218	478	1,841×	✓ Covered
EP300	chr22:41,151,813–41,152,032	220	2,800×	✓ Covered
EP300	chr22:41,152,186–41,152,370	185	7,137×	✓ Covered

FULL PANEL COVERAGE REPORT — Morrison, Toni E. (NGS-2026-052106)

Gene	chr:start–end (1-based)	Len (bp)	Depth	Coverage status
EP300	chr22:41,154,975–41,155,133	159	163×	✓ Covered
EP300	chr22:41,157,149–41,157,428	280	872×	✓ Covered
EP300	chr22:41,158,392–41,158,520	129	3,228×	✓ Covered
EP300	chr22:41,160,622–41,160,742	121	3,026×	✓ Covered
EP300	chr22:41,162,703–41,162,799	97	4,920×	✓ Covered
EP300	chr22:41,164,033–41,164,150	118	25,963×	✓ Covered
EP300	chr22:41,166,579–41,166,686	108	848×	✓ Covered
EP300	chr22:41,168,429–41,168,619	191	4,086×	✓ Covered
EP300	chr22:41,168,701–41,168,887	187	410×	✓ Covered
EP300	chr22:41,169,483–41,169,636	154	8,497×	✓ Covered
EP300	chr22:41,170,386–41,170,591	206	3,930×	✓ Covered
EP300	chr22:41,172,479–41,172,683	205	786×	✓ Covered
EP300	chr22:41,173,603–41,173,804	202	1,360×	✓ Covered
EP300	chr22:41,176,227–41,176,548	322	362×	✓ Covered
EP300	chr22:41,176,753–41,178,973	2,221	3,267×	✓ Covered
PIK3CA				
PIK3CA	chr3:179,218,190–179,218,354	165	397×	✓ Covered
PIK3CA	chr3:179,234,074–179,234,381	308	833×	✓ Covered
FBXW7				
FBXW7	chr4:152,325,986–152,326,251	266	3,502×	✓ Covered
FAT1				
FAT1	chr4:186,588,575–186,589,240	666	2,290×	✓ Covered
FAT1	chr4:186,595,669–186,595,846	178	2,277×	✓ Covered
FAT1	chr4:186,596,520–186,597,191	672	8,101×	✓ Covered
FAT1	chr4:186,597,662–186,597,812	151	1,964×	✓ Covered
FAT1	chr4:186,597,952–186,598,145	194	3,405×	✓ Covered
FAT1	chr4:186,599,878–186,600,380	503	419×	✓ Covered
FAT1	chr4:186,601,249–186,601,446	198	4,496×	✓ Covered
FAT1	chr4:186,602,883–186,603,054	172	1,035×	✓ Covered
FAT1	chr4:186,603,156–186,603,997	842	630×	✓ Covered
FAT1	chr4:186,604,357–186,604,594	238	1,935×	✓ Covered
FAT1	chr4:186,606,050–186,606,233	184	847×	✓ Covered
FAT1	chr4:186,609,163–186,609,340	178	247×	✓ Covered
FAT1	chr4:186,609,781–186,610,035	255	1,807×	✓ Covered
FAT1	chr4:186,611,366–186,611,795	430	16,119×	✓ Covered
FAT1	chr4:186,613,089–186,613,362	274	1,895×	✓ Covered
FAT1	chr4:186,614,171–186,614,364	194	13,341×	✓ Covered
FAT1	chr4:186,616,985–186,617,221	237	779×	✓ Covered
FAT1	chr4:186,617,688–186,621,795	4,108	1,188×	✓ Covered
FAT1	chr4:186,628,134–186,628,384	251	3,279×	✓ Covered
FAT1	chr4:186,628,468–186,628,783	316	3,795×	✓ Covered
FAT1	chr4:186,633,664–186,633,843	180	767×	✓ Covered
FAT1	chr4:186,636,005–186,636,255	251	3,270×	✓ Covered
FAT1	chr4:186,636,565–186,636,934	370	2,019×	✓ Covered
FAT1	chr4:186,639,702–186,639,803	102	8,060×	✓ Covered
FAT1	chr4:186,663,279–186,663,633	355	341×	✓ Covered
FAT1	chr4:186,706,543–186,709,847	3,305	1,073×	✓ Covered
TERT				
TERT	chr5:1,295,050–1,295,250	201	4,453×	✓ Covered
PIK3R1				
PIK3R1	chr5:68,294,516–68,294,698	183	4,061×	✓ Covered

FULL PANEL COVERAGE REPORT — Morrison, Toni E. (NGS-2026-052106)

Gene	chr:start–end (1-based)	Len (bp)	Depth	Coverage status
PIK3R1	chr5:68,295,128–68,295,344	217	716×	✓ Covered
PIK3R1	chr5:68,295,400–68,295,508	109	470×	✓ Covered
NSD1				
NSD1	chr5:177,135,084–177,136,050	967	1,074×	✓ Covered
NSD1	chr5:177,191,864–177,192,039	176	1,050×	✓ Covered
NSD1	chr5:177,204,100–177,204,312	213	8,075×	✓ Covered
NSD1	chr5:177,209,616–177,212,215	2,600	14,093×	✓ Covered
NSD1	chr5:177,235,801–177,235,965	165	425×	✓ Covered
NSD1	chr5:177,238,217–177,238,527	311	1,799×	✓ Covered
NSD1	chr5:177,239,736–177,239,885	150	6,355×	✓ Covered
NSD1	chr5:177,244,175–177,244,290	116	255×	✓ Covered
NSD1	chr5:177,246,658–177,246,816	159	1,229×	✓ Covered
NSD1	chr5:177,248,161–177,248,344	184	1,454×	✓ Covered
NSD1	chr5:177,251,710–177,251,873	164	2,374×	✓ Covered
NSD1	chr5:177,256,931–177,257,171	241	2,690×	✓ Covered
NSD1	chr5:177,259,969–177,260,188	220	12,098×	✓ Covered
NSD1	chr5:177,267,542–177,267,738	197	3,610×	✓ Covered
NSD1	chr5:177,269,582–177,269,827	246	964×	✓ Covered
NSD1	chr5:177,273,652–177,273,804	153	7,944×	✓ Covered
NSD1	chr5:177,280,545–177,280,854	310	370×	✓ Covered
NSD1	chr5:177,282,445–177,282,601	157	7,251×	✓ Covered
NSD1	chr5:177,283,767–177,283,948	182	1,126×	✓ Covered
NSD1	chr5:177,288,799–177,288,945	147	738×	✓ Covered
NSD1	chr5:177,291,934–177,292,178	245	7,640×	✓ Covered
NSD1	chr5:177,293,812–177,295,476	1,665	1,650×	✓ Covered
EGFR				
EGFR	chr7:55,173,901–55,174,063	163	1,032×	✓ Covered
EGFR	chr7:55,174,702–55,174,840	139	526×	✓ Covered
EGFR	chr7:55,181,273–55,181,498	226	2,317×	✓ Covered
EGFR	chr7:55,191,699–55,191,894	196	**61×**	⚠ Below 100× threshold
CDKN2A				
CDKN2A	chr9:21,968,212–21,968,262	51	4,485×	✓ Covered
CDKN2A	chr9:21,970,882–21,971,228	347	1,627×	✓ Covered
CDKN2A	chr9:21,974,658–21,974,846	189	688×	✓ Covered
CDKN2A	chr9:21,974,658–21,974,847	190	287×	✓ Covered
NOTCH1				
NOTCH1	chr9:136,496,054–136,497,578	1,525	812×	✓ Covered
NOTCH1	chr9:136,498,879–136,499,016	138	3,598×	✓ Covered
NOTCH1	chr9:136,499,092–136,499,279	188	1,615×	✓ Covered
NOTCH1	chr9:136,500,532–136,500,867	336	2,566×	✓ Covered
NOTCH1	chr9:136,501,728–136,501,933	206	999×	✓ Covered
NOTCH1	chr9:136,501,981–136,502,108	128	1,050×	✓ Covered
NOTCH1	chr9:136,502,252–136,502,508	257	1,695×	✓ Covered
NOTCH1	chr9:136,503,162–136,503,350	189	4,395×	✓ Covered
NOTCH1	chr9:136,504,653–136,505,124	472	2,840×	✓ Covered
NOTCH1	chr9:136,505,290–136,505,901	612	8,928×	✓ Covered
NOTCH1	chr9:136,506,507–136,506,659	153	8,458×	✓ Covered
NOTCH1	chr9:136,506,696–136,506,993	298	1,384×	✓ Covered
NOTCH1	chr9:136,507,285–136,507,457	173	2,981×	✓ Covered
NOTCH1	chr9:136,507,935–136,508,159	225	787×	✓ Covered
NOTCH1	chr9:136,508,212–136,508,405	194	1,524×	✓ Covered

FULL PANEL COVERAGE REPORT — Morrison, Toni E. (NGS-2026-052106)

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
NOTCH1	chr9:136,508,850–136,509,091	242	744×	✓ Covered
NOTCH1	chr9:136,509,713–136,509,981	269	635×	✓ Covered
NOTCH1	chr9:136,510,633–136,510,825	193	363×	✓ Covered
NOTCH1	chr9:136,511,132–136,511,291	160	32,371×	✓ Covered
NOTCH1	chr9:136,513,001–136,513,154	154	480×	✓ Covered
NOTCH1	chr9:136,513,372–136,513,557	186	858×	✓ Covered
NOTCH1	chr9:136,514,490–136,514,722	233	851×	✓ Covered
NOTCH1	chr9:136,515,270–136,515,420	151	**28×**	⚠ Below 100× threshold
NOTCH1	chr9:136,515,463–136,515,736	274	4,098×	✓ Covered
NOTCH1	chr9:136,515,961–136,516,114	154	920×	✓ Covered
NOTCH1	chr9:136,517,252–136,517,405	154	557×	✓ Covered
NOTCH1	chr9:136,517,732–136,517,957	226	1,706×	✓ Covered
NOTCH1	chr9:136,518,117–136,518,312	196	636×	✓ Covered
NOTCH1	chr9:136,518,571–136,518,844	274	576×	✓ Covered
NOTCH1	chr9:136,519,423–136,519,585	163	534×	✓ Covered
NOTCH1	chr9:136,522,830–136,523,208	379	2,222×	✓ Covered
NOTCH1	chr9:136,523,697–136,523,999	303	1,975×	✓ Covered
NOTCH1	chr9:136,544,004–136,544,122	119	10,444×	✓ Covered
NOTCH1	chr9:136,545,706–136,545,806	101	2,134×	✓ Covered

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CLIA: 22D2026001 | CAP: 1234567
Oral SCC Somatic Variant Panel v2.0

PATIENT & SPECIMEN INFORMATION

Patient name	Austen, Jane C.	**Accession**	NGS-2026-052104
Date of birth	1967-12-04 (Age 58)	**Specimen**	Oral swab, soft palate
MRN	40678954	**Collected**	2026-05-19
Ordering physician	Dr. Wisdom, MD	**Received**	2026-05-19
Clinical indication	Oral lesion — soft palate	**Reported**	2026-05-20

RESULT: VARIANTS DETECTED

DETECTED VARIANTS

Gene	HGVS (coding)	Protein change	VAF	Depth	Tier	Classification
EGFR	NM_005228.5:c.2573T>G	p.Leu858Arg	0.57%	5,250×	I	Pathogenic
TP53	NM_000546.6:c.524G>A	p.Arg175His	0.30%	6,690×	I	Pathogenic

VARIANT ANNOTATION

EGFR p.Leu858Arg | Pathogenic (Tier I)

EGFR p.Leu858Arg (L858R) is a well-characterised activating mutation in exon 21 of the EGFR kinase domain. It disrupts the autoinhibitory conformation, leading to constitutive EGFR kinase activation and downstream RAS/MAPK and PI3K/AKT signalling. While canonical in non-small cell lung adenocarcinoma (where it predicts response to EGFR TKIs such as osimertinib), EGFR L858R has been reported in 1–3% of OSCC. Cetuximab (anti-EGFR monoclonal antibody) is FDA-approved in HNSCC and may have enhanced activity in the setting of EGFR activating mutations. Multidisciplinary tumour board review for consideration of EGFR-directed therapy or clinical trial enrolment is recommended. Co-occurring TP53 R175H (see below) may affect response to EGFR-directed therapy and warrants combined consideration.

TP53 p.Arg175His | Pathogenic (Tier I)

TP53 p.Arg175His — see interpretation above (Dickens, Charles J., NGS-2026-052103). Co-occurrence of EGFR L858R and TP53 R175H has been associated with resistance to EGFR-directed therapies in other tumour types; multidisciplinary tumour board review is recommended. NOTE: Observed VAF (0.30%) is above the 0.2% reporting threshold and is supported by 2 consensus reads at 669× consensus depth. Orthogonal confirmation is recommended before clinical action.

GENES WITH NO PATHOGENIC OR LIKELY PATHOGENIC VARIANTS DETECTED

AJUBA, CASP8, CDKN2A, CUL3, DPYD, EP300, FAT1, FBXW7, HRAS, KEAP1, KMT2D, KRAS, NFE2L2, NOTCH1, NSD1, PIK3CA, PIK3R1, PTEN, TERT
Variants of uncertain significance (Tier III) are not reported.

METHOD SUMMARY

Cell-free and cellular DNA extracted from oral swab is subjected to hybridization capture using the IDT xGen Duplex UMI panel targeting 251 genomic regions (~77 kbp) across 21 cancer-relevant genes. Libraries incorporate 9-bp Unique Molecular Identifiers (UMIs) in a 9M1S+T read structure on both R1 and R2. Libraries are sequenced on Illumina NextSeq (2×150 bp). After alignment to GRCh38 (canonical chromosomes), reads are grouped by UMI using fgbio (adjacency strategy, ≤1 edit distance), and molecular consensus sequences are called requiring ≥2 reads per UMI family. Somatic variants are called on the consensus BAM with VarDict at minimum allele frequency 0.001. Variants ≥0.2% consensus VAF with ≥2 consensus supporting reads are reported. UMI deduplication reduces sequencing noise by >800-fold. Run-level negative controls are included on every sequencing run.

LIMITATIONS

This assay detects somatic single nucleotide variants and small indels within 251 target regions (~77 kbp) at VAF ≥0.2% with validated sensitivity of 98.4% at ≥20,000× raw input depth (~4,000× consensus after UMI deduplication). Structural variants, copy number alterations, gene fusions, large indels, intronic variants outside defined capture regions, and variants in regions with low hybridization capture efficiency are not reliably assessed. A negative result does not exclude malignancy. Tumour heterogeneity, low tumour cell content, and specimen quality may affect variant allele fraction. Variants below the 0.2% VAF reporting threshold may be present but are not reported. At 20,000× raw sequencing depth (~4,000× mean consensus), approximately 0.3% of targets may fall below 1,000× consensus depth; per-target coverage is provided in the laboratory technical report. Results should be interpreted in the context of clinical and pathological findings by a qualified clinician.

Reported by: Dr. Wisdom, MD PhD, FCAP **Report date: 2026-05-20**

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

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HOTSPOT SCREENING SUMMARY

Patient: Austen, Jane C. **Accession:** NGS-2026-052104

Hotspots listed below were explicitly interrogated in addition to all variants above 0.2 % VAF with ≥2 consensus reads across all 251 targets. Coverage criteria (≥100× consensus depth) were met at all targets.

EGFR		Cov ✓
L858R	c.2573T>G	● **DETECTED**
T790M	c.2369C>T	Not detected
Exon 19 del	c.2235_2249del	Not detected

KRAS		Cov ✓
G12D	c.35G>T	Not detected
G12V	c.35G>A	Not detected
G12C	c.34G>T	Not detected
G13D	c.38G>A	Not detected

PIK3CA		Cov ✓
E542K	c.1624G>A	Not detected
E545K	c.1633G>A	Not detected
H1047R	c.3140A>G	Not detected

TERT		Cov ✓
C228T	c.-124C>T	Not detected
C250T	c.-146C>T	Not detected

HRAS		Cov ✓
G12V	c.35G>T	Not detected
G12D	c.35G>A	Not detected
Q61L	c.182A>T	Not detected
Q61R	c.182A>G	Not detected

NOTCH1		Cov ✓
P2415L	c.7244C>T	Not detected
R1699W	c.5095C>T	Not detected

TP53		Cov ✓
R175H	c.524G>A	● **DETECTED**
R248Q	c.743G>A	Not detected
R248W	c.742C>T	Not detected
R273H	c.818G>A	Not detected
R282W	c.844C>T	Not detected

REMAINING PANEL GENES — SEQUENCE-LEVEL DETECTION, NO PREDEFINED HOTSPOT LIST

AJUBA · CASP8 · CDKN2A · CUL3 · DPYD · EP300 · FAT1 · FBXW7 · KEAP1 · KMT2D · NFE2L2 · NSD1 · PIK3R1 · PTEN

All variants above 0.2 % VAF with ≥2 consensus reads are reported for these genes. No variants meeting reporting criteria were detected. All targets in these genes met minimum coverage criteria (≥100× consensus depth).

Reported by: Dr. Wisdom, MD PhD, FCAP **Report date: 2026-05-20**

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

FULL PANEL COVERAGE REPORT — Austen, Jane C. (NGS-2026-052104)

251 targets, 77,417 bp total. Mean consensus depth: 4,040×. Targets below 100× consensus: 0 / 251 (0%). Depths are derived from the consensus BAM (UMI deduplication, min-reads ≥ 2). Positions reported 1-based.

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
DPYD				
DPYD	chr1:97,082,291–97,082,491	201	2,767×	✓ Covered
DPYD	chr1:97,449,958–97,450,158	201	5,001×	✓ Covered
DPYD	chr1:97,515,687–97,515,887	201	821×	✓ Covered
DPYD	chr1:97,573,763–97,573,963	201	4,374×	✓ Covered
PTEN				
PTEN	chr10:87,864,450–87,864,568	119	1,577×	✓ Covered
PTEN	chr10:87,894,005–87,894,129	125	1,888×	✓ Covered
PTEN	chr10:87,925,493–87,925,577	85	335×	✓ Covered
PTEN	chr10:87,931,026–87,931,109	84	8,450×	✓ Covered
PTEN	chr10:87,932,993–87,933,271	279	2,486×	✓ Covered
PTEN	chr10:87,952,098–87,952,279	182	2,839×	✓ Covered
PTEN	chr10:87,957,833–87,958,039	207	645×	✓ Covered
PTEN	chr10:87,960,874–87,961,138	265	2,489×	✓ Covered
PTEN	chr10:87,965,267–87,965,489	223	6,186×	✓ Covered
HRAS				
HRAS	chr11:532,619–532,775	157	667×	✓ Covered
HRAS	chr11:533,433–533,632	200	279×	✓ Covered
HRAS	chr11:533,746–533,964	219	3,247×	✓ Covered
HRAS	chr11:534,192–534,342	151	3,293×	✓ Covered
KRAS				
KRAS	chr12:25,209,778–25,209,931	154	1,191×	✓ Covered
KRAS	chr12:25,225,594–25,225,793	200	3,312×	✓ Covered
KRAS	chr12:25,227,214–25,227,432	219	6,076×	✓ Covered
KRAS	chr12:25,245,254–25,245,404	151	1,825×	✓ Covered
KMT2D				
KMT2D	chr12:49,021,763–49,021,892	130	4,312×	✓ Covered
KMT2D	chr12:49,022,023–49,022,171	149	1,020×	✓ Covered
KMT2D	chr12:49,022,260–49,022,373	114	180×	✓ Covered
KMT2D	chr12:49,022,570–49,022,895	326	1,531×	✓ Covered
KMT2D	chr12:49,024,558–49,024,728	171	926×	✓ Covered
KMT2D	chr12:49,024,790–49,024,966	177	3,423×	✓ Covered
KMT2D	chr12:49,026,162–49,027,342	1,181	2,510×	✓ Covered
KMT2D	chr12:49,027,783–49,027,950	168	1,089×	✓ Covered
KMT2D	chr12:49,027,989–49,028,161	173	1,960×	✓ Covered
KMT2D	chr12:49,028,808–49,028,978	171	16,328×	✓ Covered
KMT2D	chr12:49,029,041–49,029,256	216	747×	✓ Covered
KMT2D	chr12:49,029,381–49,029,496	116	1,451×	✓ Covered
KMT2D	chr12:49,030,260–49,030,459	200	5,009×	✓ Covered
KMT2D	chr12:49,030,581–49,030,788	208	14,902×	✓ Covered
KMT2D	chr12:49,030,873–49,031,053	181	8,313×	✓ Covered
KMT2D	chr12:49,031,155–49,033,984	2,830	1,416×	✓ Covered
KMT2D	chr12:49,034,047–49,034,319	273	1,247×	✓ Covered
KMT2D	chr12:49,034,390–49,034,496	107	1,226×	✓ Covered
KMT2D	chr12:49,034,562–49,034,686	125	4,825×	✓ Covered
KMT2D	chr12:49,034,792–49,034,955	164	2,267×	✓ Covered
KMT2D	chr12:49,037,105–49,039,009	1,905	2,586×	✓ Covered
KMT2D	chr12:49,039,202–49,039,378	177	14,093×	✓ Covered
KMT2D	chr12:49,039,415–49,039,637	223	1,723×	✓ Covered

FULL PANEL COVERAGE REPORT — Austen, Jane C. (NGS-2026-052104)

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KMT2D	chr12:49,039,704–49,041,555	1,852	1,674×	✓ Covered
KMT2D	chr12:49,041,635–49,041,725	91	1,174×	✓ Covered
KMT2D	chr12:49,041,897–49,042,010	114	2,527×	✓ Covered
KMT2D	chr12:49,042,069–49,042,350	282	1,007×	✓ Covered
KMT2D	chr12:49,042,541–49,042,665	125	2,051×	✓ Covered
KMT2D	chr12:49,042,721–49,042,898	178	871×	✓ Covered
KMT2D	chr12:49,043,056–49,043,206	151	1,219×	✓ Covered
KMT2D	chr12:49,043,343–49,043,448	106	1,425×	✓ Covered
KMT2D	chr12:49,043,615–49,043,802	188	505×	✓ Covered
KMT2D	chr12:49,043,848–49,044,018	171	1,534×	✓ Covered
KMT2D	chr12:49,044,180–49,044,324	145	21,559×	✓ Covered
KMT2D	chr12:49,044,383–49,044,542	160	367×	✓ Covered
KMT2D	chr12:49,044,724–49,044,985	262	1,256×	✓ Covered
KMT2D	chr12:49,045,900–49,045,987	88	1,688×	✓ Covered
KMT2D	chr12:49,046,045–49,046,194	150	2,907×	✓ Covered
KMT2D	chr12:49,046,240–49,046,444	205	996×	✓ Covered
KMT2D	chr12:49,046,589–49,046,810	222	858×	✓ Covered
KMT2D	chr12:49,047,945–49,048,089	145	1,516×	✓ Covered
KMT2D	chr12:49,048,639–49,048,789	151	2,230×	✓ Covered
KMT2D	chr12:49,049,085–49,049,238	154	3,242×	✓ Covered
KMT2D	chr12:49,049,662–49,050,810	1,149	5,430×	✓ Covered
KMT2D	chr12:49,050,866–49,052,444	1,579	857×	✓ Covered
KMT2D	chr12:49,052,544–49,052,729	186	672×	✓ Covered
KMT2D	chr12:49,052,895–49,053,092	198	2,100×	✓ Covered
KMT2D	chr12:49,053,187–49,053,341	155	633×	✓ Covered
KMT2D	chr12:49,053,456–49,053,661	206	6,125×	✓ Covered
KMT2D	chr12:49,053,958–49,054,160	203	1,542×	✓ Covered
KMT2D	chr12:49,054,287–49,054,436	150	498×	✓ Covered
KMT2D	chr12:49,054,508–49,054,771	264	1,132×	✓ Covered
KMT2D	chr12:49,054,880–49,055,046	167	2,301×	✓ Covered
KMT2D	chr12:49,055,256–49,055,344	89	1,443×	✓ Covered
AJUBA				
AJUBA	chr14:22,973,426–22,973,588	163	55,542×	✓ Covered
AJUBA	chr14:22,974,027–22,974,135	109	879×	✓ Covered
AJUBA	chr14:22,974,819–22,974,910	92	3,131×	✓ Covered
AJUBA	chr14:22,974,954–22,975,124	171	2,771×	✓ Covered
AJUBA	chr14:22,976,436–22,976,538	103	1,078×	✓ Covered
AJUBA	chr14:22,976,625–22,976,732	108	2,457×	✓ Covered
AJUBA	chr14:22,978,324–22,978,465	142	3,463×	✓ Covered
AJUBA	chr14:22,981,241–22,982,286	1,046	701×	✓ Covered
TP53				
TP53	chr17:7,669,592–7,669,710	119	1,032×	✓ Covered
TP53	chr17:7,670,589–7,670,735	147	1,641×	✓ Covered
TP53	chr17:7,673,515–7,673,628	114	20,053×	✓ Covered
TP53	chr17:7,673,681–7,673,857	177	7,504×	✓ Covered
TP53	chr17:7,674,161–7,674,310	150	4,211×	✓ Covered
TP53	chr17:7,674,839–7,674,991	153	1,760×	✓ Covered
TP53	chr17:7,675,033–7,675,256	224	1,434×	● **VARIANT DETECTED**
TP53	chr17:7,675,974–7,676,292	319	1,851×	✓ Covered
TP53	chr17:7,676,362–7,676,423	62	3,457×	✓ Covered
TP53	chr17:7,676,501–7,676,614	114	16,258×	✓ Covered

FULL PANEL COVERAGE REPORT — Austen, Jane C. (NGS-2026-052104)

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KEAP1				
KEAP1	chr19:10,486,635–10,486,838	204	9,195×	✓ Covered
KEAP1	chr19:10,489,172–10,489,388	217	2,498×	✓ Covered
KEAP1	chr19:10,489,628–10,489,873	246	3,054×	✓ Covered
KEAP1	chr19:10,491,557–10,492,282	726	301×	✓ Covered
KEAP1	chr19:10,499,375–10,500,053	679	2,440×	✓ Covered
NFE2L2				
NFE2L2	chr2:177,233,985–177,234,291	307	379×	✓ Covered
CASP8				
CASP8	chr2:201,266,467–201,266,811	345	699×	✓ Covered
CASP8	chr2:201,271,496–201,271,641	146	428×	✓ Covered
CASP8	chr2:201,272,618–201,272,796	179	1,547×	✓ Covered
CASP8	chr2:201,272,878–201,272,962	85	1,965×	✓ Covered
CASP8	chr2:201,274,869–201,274,973	105	5,441×	✓ Covered
CASP8	chr2:201,276,807–201,276,988	182	1,705×	✓ Covered
CASP8	chr2:201,284,796–201,285,337	542	2,381×	✓ Covered
CASP8	chr2:201,286,439–201,286,611	173	940×	✓ Covered
CUL3				
CUL3	chr2:224,474,228–224,474,396	169	1,646×	✓ Covered
CUL3	chr2:224,478,180–224,478,365	186	1,036×	✓ Covered
CUL3	chr2:224,481,872–224,482,098	227	2,732×	✓ Covered
CUL3	chr2:224,495,812–224,495,986	175	883×	✓ Covered
CUL3	chr2:224,497,733–224,497,869	137	6,278×	✓ Covered
CUL3	chr2:224,500,343–224,500,507	165	2,270×	✓ Covered
CUL3	chr2:224,502,945–224,503,092	148	1,317×	✓ Covered
CUL3	chr2:224,503,632–224,503,842	211	1,348×	✓ Covered
CUL3	chr2:224,505,936–224,506,152	217	8,166×	✓ Covered
CUL3	chr2:224,506,838–224,507,023	186	2,405×	✓ Covered
CUL3	chr2:224,511,334–224,511,602	269	5,100×	✓ Covered
CUL3	chr2:224,513,504–224,513,658	155	7,531×	✓ Covered
CUL3	chr2:224,514,592–224,514,792	201	5,838×	✓ Covered
CUL3	chr2:224,535,508–224,535,661	154	1,233×	✓ Covered
CUL3	chr2:224,557,639–224,557,876	238	6,636×	✓ Covered
CUL3	chr2:224,584,924–224,585,029	106	1,357×	✓ Covered
EP300				
EP300	chr22:41,092,985–41,093,118	134	12,660×	✓ Covered
EP300	chr22:41,117,167–41,117,841	675	977×	✓ Covered
EP300	chr22:41,125,844–41,126,060	217	754×	✓ Covered
EP300	chr22:41,127,467–41,127,768	302	444×	✓ Covered
EP300	chr22:41,129,870–41,130,023	154	2,472×	✓ Covered
EP300	chr22:41,131,368–41,131,653	286	4,908×	✓ Covered
EP300	chr22:41,135,793–41,135,926	134	1,864×	✓ Covered
EP300	chr22:41,137,633–41,137,810	178	18,633×	✓ Covered
EP300	chr22:41,140,120–41,140,277	158	2,006×	✓ Covered
EP300	chr22:41,141,028–41,141,242	215	521×	✓ Covered
EP300	chr22:41,146,719–41,146,836	118	24,464×	✓ Covered
EP300	chr22:41,147,817–41,147,966	150	920×	✓ Covered
EP300	chr22:41,149,018–41,149,195	178	9,560×	✓ Covered
EP300	chr22:41,149,741–41,150,218	478	776×	✓ Covered
EP300	chr22:41,151,813–41,152,032	220	415×	✓ Covered
EP300	chr22:41,152,186–41,152,370	185	447×	✓ Covered

FULL PANEL COVERAGE REPORT — Austen, Jane C. (NGS-2026-052104)

Gene	chr:start–end (1-based)	Len (bp)	Depth	Coverage status
EP300	chr22:41,154,975–41,155,133	159	1,144×	✓ Covered
EP300	chr22:41,157,149–41,157,428	280	3,876×	✓ Covered
EP300	chr22:41,158,392–41,158,520	129	8,397×	✓ Covered
EP300	chr22:41,160,622–41,160,742	121	13,851×	✓ Covered
EP300	chr22:41,162,703–41,162,799	97	3,885×	✓ Covered
EP300	chr22:41,164,033–41,164,150	118	626×	✓ Covered
EP300	chr22:41,166,579–41,166,686	108	1,209×	✓ Covered
EP300	chr22:41,168,429–41,168,619	191	268×	✓ Covered
EP300	chr22:41,168,701–41,168,887	187	580×	✓ Covered
EP300	chr22:41,169,483–41,169,636	154	1,747×	✓ Covered
EP300	chr22:41,170,386–41,170,591	206	1,515×	✓ Covered
EP300	chr22:41,172,479–41,172,683	205	8,062×	✓ Covered
EP300	chr22:41,173,603–41,173,804	202	2,160×	✓ Covered
EP300	chr22:41,176,227–41,176,548	322	772×	✓ Covered
EP300	chr22:41,176,753–41,178,973	2,221	1,190×	✓ Covered
PIK3CA				
PIK3CA	chr3:179,218,190–179,218,354	165	2,168×	✓ Covered
PIK3CA	chr3:179,234,074–179,234,381	308	294×	✓ Covered
FBXW7				
FBXW7	chr4:152,325,986–152,326,251	266	1,675×	✓ Covered
FAT1				
FAT1	chr4:186,588,575–186,589,240	666	541×	✓ Covered
FAT1	chr4:186,595,669–186,595,846	178	13,585×	✓ Covered
FAT1	chr4:186,596,520–186,597,191	672	3,088×	✓ Covered
FAT1	chr4:186,597,662–186,597,812	151	1,461×	✓ Covered
FAT1	chr4:186,597,952–186,598,145	194	13,495×	✓ Covered
FAT1	chr4:186,599,878–186,600,380	503	1,003×	✓ Covered
FAT1	chr4:186,601,249–186,601,446	198	1,590×	✓ Covered
FAT1	chr4:186,602,883–186,603,054	172	434×	✓ Covered
FAT1	chr4:186,603,156–186,603,997	842	2,156×	✓ Covered
FAT1	chr4:186,604,357–186,604,594	238	1,126×	✓ Covered
FAT1	chr4:186,606,050–186,606,233	184	575×	✓ Covered
FAT1	chr4:186,609,163–186,609,340	178	6,755×	✓ Covered
FAT1	chr4:186,609,781–186,610,035	255	6,848×	✓ Covered
FAT1	chr4:186,611,366–186,611,795	430	5,319×	✓ Covered
FAT1	chr4:186,613,089–186,613,362	274	3,749×	✓ Covered
FAT1	chr4:186,614,171–186,614,364	194	813×	✓ Covered
FAT1	chr4:186,616,985–186,617,221	237	6,938×	✓ Covered
FAT1	chr4:186,617,688–186,621,795	4,108	820×	✓ Covered
FAT1	chr4:186,628,134–186,628,384	251	888×	✓ Covered
FAT1	chr4:186,628,468–186,628,783	316	3,101×	✓ Covered
FAT1	chr4:186,633,664–186,633,843	180	1,063×	✓ Covered
FAT1	chr4:186,636,005–186,636,255	251	1,318×	✓ Covered
FAT1	chr4:186,636,565–186,636,934	370	671×	✓ Covered
FAT1	chr4:186,639,702–186,639,803	102	2,920×	✓ Covered
FAT1	chr4:186,663,279–186,663,633	355	2,017×	✓ Covered
FAT1	chr4:186,706,543–186,709,847	3,305	5,212×	✓ Covered
TERT				
TERT	chr5:1,295,050–1,295,250	201	7,348×	✓ Covered
PIK3R1				
PIK3R1	chr5:68,294,516–68,294,698	183	4,473×	✓ Covered

FULL PANEL COVERAGE REPORT — Austen, Jane C. (NGS-2026-052104)

Gene	chr:start–end (1-based)	Len (bp)	Depth	Coverage status
PIK3R1	chr5:68,295,128–68,295,344	217	15,438×	✓ Covered
PIK3R1	chr5:68,295,400–68,295,508	109	9,171×	✓ Covered
NSD1				
NSD1	chr5:177,135,084–177,136,050	967	6,809×	✓ Covered
NSD1	chr5:177,191,864–177,192,039	176	816×	✓ Covered
NSD1	chr5:177,204,100–177,204,312	213	3,398×	✓ Covered
NSD1	chr5:177,209,616–177,212,215	2,600	2,875×	✓ Covered
NSD1	chr5:177,235,801–177,235,965	165	4,375×	✓ Covered
NSD1	chr5:177,238,217–177,238,527	311	3,622×	✓ Covered
NSD1	chr5:177,239,736–177,239,885	150	2,110×	✓ Covered
NSD1	chr5:177,244,175–177,244,290	116	2,636×	✓ Covered
NSD1	chr5:177,246,658–177,246,816	159	695×	✓ Covered
NSD1	chr5:177,248,161–177,248,344	184	5,977×	✓ Covered
NSD1	chr5:177,251,710–177,251,873	164	1,627×	✓ Covered
NSD1	chr5:177,256,931–177,257,171	241	3,435×	✓ Covered
NSD1	chr5:177,259,969–177,260,188	220	21,546×	✓ Covered
NSD1	chr5:177,267,542–177,267,738	197	2,440×	✓ Covered
NSD1	chr5:177,269,582–177,269,827	246	13,753×	✓ Covered
NSD1	chr5:177,273,652–177,273,804	153	4,199×	✓ Covered
NSD1	chr5:177,280,545–177,280,854	310	1,423×	✓ Covered
NSD1	chr5:177,282,445–177,282,601	157	3,452×	✓ Covered
NSD1	chr5:177,283,767–177,283,948	182	4,757×	✓ Covered
NSD1	chr5:177,288,799–177,288,945	147	2,777×	✓ Covered
NSD1	chr5:177,291,934–177,292,178	245	3,599×	✓ Covered
NSD1	chr5:177,293,812–177,295,476	1,665	4,800×	✓ Covered
EGFR				
EGFR	chr7:55,173,901–55,174,063	163	2,953×	✓ Covered
EGFR	chr7:55,174,702–55,174,840	139	1,810×	✓ Covered
EGFR	chr7:55,181,273–55,181,498	226	6,729×	✓ Covered
EGFR	chr7:55,191,699–55,191,894	196	2,314×	● **VARIANT DETECTED**
CDKN2A				
CDKN2A	chr9:21,968,212–21,968,262	51	4,826×	✓ Covered
CDKN2A	chr9:21,970,882–21,971,228	347	18,659×	✓ Covered
CDKN2A	chr9:21,974,658–21,974,846	189	7,274×	✓ Covered
CDKN2A	chr9:21,974,658–21,974,847	190	1,321×	✓ Covered
NOTCH1				
NOTCH1	chr9:136,496,054–136,497,578	1,525	3,992×	✓ Covered
NOTCH1	chr9:136,498,879–136,499,016	138	843×	✓ Covered
NOTCH1	chr9:136,499,092–136,499,279	188	608×	✓ Covered
NOTCH1	chr9:136,500,532–136,500,867	336	6,156×	✓ Covered
NOTCH1	chr9:136,501,728–136,501,933	206	2,752×	✓ Covered
NOTCH1	chr9:136,501,981–136,502,108	128	5,048×	✓ Covered
NOTCH1	chr9:136,502,252–136,502,508	257	7,721×	✓ Covered
NOTCH1	chr9:136,503,162–136,503,350	189	18,277×	✓ Covered
NOTCH1	chr9:136,504,653–136,505,124	472	2,436×	✓ Covered
NOTCH1	chr9:136,505,290–136,505,901	612	2,309×	✓ Covered
NOTCH1	chr9:136,506,507–136,506,659	153	961×	✓ Covered
NOTCH1	chr9:136,506,696–136,506,993	298	699×	✓ Covered
NOTCH1	chr9:136,507,285–136,507,457	173	2,438×	✓ Covered
NOTCH1	chr9:136,507,935–136,508,159	225	903×	✓ Covered
NOTCH1	chr9:136,508,212–136,508,405	194	808×	✓ Covered

FULL PANEL COVERAGE REPORT — Austen, Jane C. (NGS-2026-052104)

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
NOTCH1	chr9:136,508,850–136,509,091	242	1,771×	✓ Covered
NOTCH1	chr9:136,509,713–136,509,981	269	411×	✓ Covered
NOTCH1	chr9:136,510,633–136,510,825	193	7,140×	✓ Covered
NOTCH1	chr9:136,511,132–136,511,291	160	13,495×	✓ Covered
NOTCH1	chr9:136,513,001–136,513,154	154	56,236×	✓ Covered
NOTCH1	chr9:136,513,372–136,513,557	186	1,111×	✓ Covered
NOTCH1	chr9:136,514,490–136,514,722	233	1,606×	✓ Covered
NOTCH1	chr9:136,515,270–136,515,420	151	3,284×	✓ Covered
NOTCH1	chr9:136,515,463–136,515,736	274	956×	✓ Covered
NOTCH1	chr9:136,515,961–136,516,114	154	4,534×	✓ Covered
NOTCH1	chr9:136,517,252–136,517,405	154	2,223×	✓ Covered
NOTCH1	chr9:136,517,732–136,517,957	226	2,568×	✓ Covered
NOTCH1	chr9:136,518,117–136,518,312	196	311×	✓ Covered
NOTCH1	chr9:136,518,571–136,518,844	274	1,304×	✓ Covered
NOTCH1	chr9:136,519,423–136,519,585	163	1,767×	✓ Covered
NOTCH1	chr9:136,522,830–136,523,208	379	6,330×	✓ Covered
NOTCH1	chr9:136,523,697–136,523,999	303	186×	✓ Covered
NOTCH1	chr9:136,544,004–136,544,122	119	815×	✓ Covered
NOTCH1	chr9:136,545,706–136,545,806	101	1,827×	✓ Covered